

Unified Consciousness Theory I

Structural-Experiential Identity and the Continuity from
Physical Process to Conceptual Self

Hongju Liu

Independent researcher, Shenzhen, China

Report: TA-TR-2026-20
Version: 1.2
Date: 4 October 2026
Prior published versions: v1.0, DOI 10.5281/zenodo.23005588
v1.1, DOI 10.5281/zenodo.23030207
v1.2 DOI: 10.5281/zenodo.23131575

Contents

Abstract	1
1. The target problem	2
2. Ontological domain: actual process tokens	2
2.1 Actuality and tokenhood	2
2.2 Multiscale token plurality	3
2.3 P6 — Token, type, and lineage	3
2.4 P3 — Persistence, cessation, and embedding	4
2.5 A concrete finite process criterion	4
3. Ontic organization, views, estimates, and subjects	5
3.1 Full token-relative ontic organization	5
3.2 Physical scientific views	5
3.3 Scientific estimates	5
3.4 Subjecthood	6
3.5 A pointed, port-aware comparison signature	6
3.6 Structural-type space and invariant topology	7
4. One consciousness-specific core commitment	7
C1 — Structural-Experiential Identity	7
Lemma C1-OI — Pairwise type equivalence	8
Lemma C1-W — One-way phenomenal completeness	8
Theorem U1 — Universal Nonempty Experience	8
Theorem U2 — Structural Continuity	8
Corollary U3 — Selfhood Non-Prerequisite	9
Migration table	9
Proposition U0 — Foundational Compression	10
5. Formal consequences	10
5.1 Organization-Gate Impossibility	10
5.2 No Hidden Quale	11
5.3 Probe Invariance	11
5.4 No Future Contamination	11
5.5 Causally Screened History / No Ghost History	12
5.6 External reward-label non-identity	12
6. Factorization, macroprocesses, and combination	12
6.1 Independent products and comparison objects	12
6.2 Why C1-OI alone is insufficient	12
6.3 Non-summative combination from the intended identity	13
6.4 Macro invariance with micro change	14
7. Subject selection under symmetry: an orbit obstruction	14
7.1 Equivariant selection is a union of candidate orbits	14
7.2 The three-part example and its counterlimits	15
8. No first conscious ancestor: from physical process to conceptual self	15
8.1 Corollary E1 — No First Conscious Ancestor	15
8.2 The All-Ancestors-Survive Thought Experiment	16
Corollary E2 — Fine-Grained Structural Chains	17
Explanatory pressure on ignition theories	18
8.3 Why Evolutionary Continuity Alone Does Not Prove UCT	18
Conditional Lemma E3 — Connected Existence Constancy	18
8.4 Elementary physical processes	19
8.5 From stable chemistry to self-maintaining organization	20
8.6 Single-cell and non-neural learning: temporal depth	20
8.7 Multicellularity: from juxtaposition to organism-scale relations	20

8.8 Nervous systems: routing and specialization, not an experience switch	21
8.9 Damage regulation, nociception, and affect	21
One explanatory principle across experiential qualities	22
8.10 Association, relational abstraction, and reasoning	22
8.11 Physical witness and semantic inflation	22
8.12 Body-centered organization	23
8.13 Self-model	23
8.14 The conceptual “I”	23
8.15 No universal consciousness ladder	24
9. Structural transformation versus enrichment	24
9.1 An exact edge-addition family with unchanged declared output	25
9.2 Proposition: three distinct causal-retention regimes	26
9.3 Conditional experiential interpretation	26
9.4 Shared physical example, not an independent proof of identity	26
10. Scientific bridge architecture	27
10.1 Declared phenomenal target	27
10.2 Sufficiency and reflection are different bridge hypotheses	27
10.3 Operational Structural Correspondence	27
11. Physical Token/View Selection Protocol	28
12. Empirical failure conditions	28
12.1 Same frozen physical view, different phenomenal target	28
12.2 Different frozen physical views, identical phenomenal target	29
12.3 Behaviorally matched causal rewiring	29
12.4 Latent-memory test	30
13. Relation to UCT II and neighboring theories	30
14. Strong objections and limitations	33
14.1 The core identity commitment and process ontology are strong	33
14.2 C1 risks tautological explanatory rhetoric	33
14.3 Complete ontic organization is not directly observed	33
14.4 Complete phenomenology is not directly measured	34
14.5 Process proliferation	34
14.6 Macroprocess descriptions are applications, not new existence gates	34
14.7 Subject terminology must not become a prerequisite	34
14.8 General quality identity and biological specification	34
14.9 Cross-species qualia identity is underidentified	35
14.10 Identity, isomorphism, and explanatory strength	35
14.11 Metaphysical structuralism objections remain substantive	35
14.12 Russellian and neutral-structuralist alternatives	35
14.13 Topology underdetermination	36
14.14 Relativistic and quantum extensions	36
14.15 Semantic representation	36
14.16 Axiom minimization is not independent evidence	36
14.17 Formal strength and finite measurements must remain explicit	37
15. Claim-status summary	37
16. Conclusion	38
16.1 Cross-version compatibility with UCT II v1.1	39
References	40
Appendix A. Main changes relative to v1.1	42
Appendix B. An exact three-register thought experiment	43

B.1 Physical specification and scope	43
B.2 Exact closure of an implemented sub-process	44
B.3 Full structural difference is invariant to state relabeling	45
B.4 Two checks against arbitrary coarse-graining	45
B.5 What the example does and does not distinguish	46
B.6 Exact verification record	46
Appendix C. Revision-specific dependency ledger	46
C.1 Formal-DAG audit	48
C.2 Conceptual firewall map	49
Appendix D. Optional checks on a finite coarse description	49

Versioning statement

This manuscript is version 1.2, succeeding *Unified Consciousness Theory I* v1.1 (DOI 10.5281/zenodo.23030207). It does not overwrite or retroactively alter v1.0 or v1.1. The principal change is architectural rather than empirical: the v1.1 foundational list is reclassified according to logical role. Persistence under embedding and token/type/lineage move to process ontology; Universal Experience and Selfhood Non-Prerequisite become derived results; Scoped Structural Continuity is derived after placing physical and experiential organization in a shared isomorphism-class space with a predeclared invariant topology; the strong tokenwise Structural-Experiential Identity remains the single consciousness-specific core axiom. Evolutionary continuity remains an explanatory application and threshold-pressure argument, not a proof of the core identity principle.

AI assistance disclosure

Substantial ChatGPT (OpenAI GPT-5.6 Sol) assistance was used for literature/source retrieval, formalization, adversarial review, dependency auditing, theorem checking, code-based static checks, drafting, and editing under human direction. The human author of record is responsible for the theory commitments and any publication decision. Version 1.2 additionally used ChatGPT to minimize independent commitments, formalize the structural quotient-space continuity result, migrate theorem dependencies, and construct cross-version compatibility tables. These checks are not independent peer review; no separate final human line-by-line verification is claimed.

Abstract

Unified Consciousness Theory (UCT) is reformulated around one consciousness-specific commitment: for every actual valid physical process token, its complete token-relative physical organization and its intrinsic experiential organization are the same structure under a declared common signature. Process ontology remains separate and explicit, covering actual support, causal continuity, boundary accountability, persistence, lineage, and token/type distinctions.

This architecture makes several v1.1 commitments derivable rather than independent. Universal nonempty experience (U1) follows from the nonempty constitutive organization of an actual token together with tokenwise Structural-Experiential Identity (C1). Structural continuity (U2) follows when physical and experiential descriptions are represented in the same isomorphism-class space with a predeclared invariant topology or pseudometric. Selfhood Non-Prerequisite becomes a corollary given an actual non-self token witness, while persistence under embedding and token/type/lineage are treated as process-ontological rules.

The reformulation preserves the main v1.1 consequences concerning organization-gate impossibility, hidden-quale exclusion under fixed complete structure, probe invariance, no future contamination, causally screened history, non-summative combination, subject-selection symmetry, structural transformation, and failure-prone finite bridge methodology. It also retains the distinction between complete ontic organization, finite physical views, scientific estimates, and subject individuation.

A central evolutionary consequence is explicit: within UCT there is no first conscious ancestor. U1 assigns nonempty experience to every actual valid process token, including the simplest actual physical interaction processes. Evolution may introduce thresholds for memory, recurrence, global access, self-modeling, language, metacognition, and conceptual selfhood, but these are transitions within experiential organization rather than an ignition of experience itself. U2 transfers strict continuity when a physical structural path is continuous, while a fine-chain corollary handles arbitrarily gradual change without requiring a perfectly smooth global trajectory. Evolutionary continuity is therefore an application and explanatory pressure test, not an independent proof of C1.

The revision reduces logical redundancy but does not reduce the empirical burden. C1 remains a strong metaphysical identity commitment; finite experiments test explicitly declared bridge packages rather than complete ontic identity directly.

Keywords: consciousness; process ontology; structural identity; axiom minimization; continuity; evolution; experiential structure; self-model; scientific views

1. The target problem

Contemporary consciousness theories assign central roles to integrated causal structure, recurrent processing, global availability, higher-order representation, predictive modeling, information closure, temporal organization, and related mechanisms (Seth & Bayne, 2022; Albantakis et al., 2023; Chang et al., 2020; Kanai & Fujisawa, 2024). Much of the competition among theories concerns whether one of these mechanisms constitutes, enables, or marks consciousness.

UCT begins by separating two questions:

Does experience exist?

and:

What is the organization of that experience?

UCT no longer treats the first question as an independent axiom. Instead, both existence and organization are placed under one tokenwise structural-experiential identity commitment. Given an actual token with nonempty physical organization, nonempty experience follows as a theorem. The nontrivial scientific problem is then to identify and compare organization without confusing finite views with complete ontic structure.

The single consciousness-specific starting point is tokenwise structural-experiential identity. Universal experience is a direct internal consequence, while process identity and persistence are handled in the process ontology. Internal derivations are assessed by whether they follow from those commitments. The substantive comparative task is to constrain structured differences, multiscale relations, and transformations; empirical applications additionally require declared bridge predictions. Human and cellular experience do not require different existence mechanisms within this framework.

The principal clarification of version 1.2 is that the strong identity commitment makes part of the v1.1 axiom list logically redundant. The second is that continuity must be stated in a common structural-type space, not by silently assigning unrelated topologies to physical and experiential descriptions. A central evolutionary consequence is then explicit: within UCT there is no first conscious ancestor. Thresholds may occur for particular forms of organization, but not for the existence of experience itself. Finite experiments still do not directly observe complete ontic organization or complete phenomenology; operational tests therefore target explicitly declared finite bridge hypotheses.

2. Ontological domain: actual process tokens

2.1 Actuality and tokenhood

Let:

$$\mathcal{P}$$

be the domain of actual valid physical process tokens. An element:

$$P \in \mathcal{P}$$

is an actual causally continuous physical subhistory, not an arbitrary set of variables selected after the fact.

A valid token must satisfy, at minimum:

1. **actual support** — the relevant states/events are physically instantiated;
2. **induced-relation fidelity** — attributed internal relations are inherited from physical organization rather than invented by the modeller;
3. **causal continuity** — the token has a physically traceable causal history over the interval of interest;
4. **boundary accountability** — materially relevant cross-boundary influences are represented rather than silently erased;
5. **lineage traceability** — continuation, replacement, branching, merger, and cessation are physically trackable;
6. **description invariance** — invertible coordinate or label changes do not create new tokens.

UCT is process-token pluralist. Actual tokens may:

- overlap;
- nest;
- branch;
- merge;
- persist through replacement;
- exist at multiple physical scales.

No maximality or exclusion rule is imposed at the basal level.

2.2 Multiscale token plurality

A critical distinction in v1.1 is:

actual macroprocess token $\neq$ arbitrary coarse description.

Molecular, cellular, circuit-level, neural-system, organism-level, and other macroprocesses may all be actual process tokens when the relevant organization is physically instantiated. A scientist does not create a macroprocess merely by averaging microscopic variables.

Macroprocess is a scale-relative description of actual organized physical activity. Its experience does not wait for a new threshold of autonomy, recurrence, closure, complexity, or subjecthood. Many overlapping and nested processes may qualify; UCT does not require a unique privileged macro boundary. In an application, one must specify which actual causal organization is being described, just as one must specify which physical process a model concerns. That identification task should not be confused with a missing mechanism that turns macroprocesses into experience.

2.3 P6 — Token, type, and lineage

UCT distinguishes:

structural type $\neq$ numerical token $\neq$ causal lineage.

Two process tokens can have isomorphic organization without being numerically identical:

$$\mathbf{D}(P) \cong \mathbf{D}(Q) \not\Rightarrow P = Q.$$

Lineage is a directed continuation relation rather than numerical identity. A process may branch:

$$P \rightsquigarrow Q_1, \quad P \rightsquigarrow Q_2, \quad Q_1 \neq Q_2,$$

or contribute to a merger:

$$P \rightsquigarrow M, \quad Q \rightsquigarrow M.$$

Nothing in these relations alone determines a unique subject count.

2.4 P3 — Persistence, cessation, and embedding

Embedding is not by itself a termination event. If a process P continues to satisfy the actual-token criterion over interval I while participating in a larger actual process Q , then:

$$\boxed{Persist(P, I) \wedge Embed(P, Q) \Rightarrow P \in \mathcal{P} \text{ throughout } I.}$$

Whether persistence actually holds is a physical question concerning support, constitutive dynamics, boundary dependence, and lineage. This rule does not assert that a process survives after its constitutive dynamics cease.

The distinctive role formerly assigned to A3 is therefore process ontological. Once persistence establishes that P remains an actual token, the experience conclusion follows from Theorem U1 below.

2.5 A concrete finite process criterion

In a finite classical description, let G represent physically instantiated events and their causal relations. An admissible process subhistory H consists of actual events, inherits their physical relations, is connected through the inherited causal links over the declared interval, and records materially relevant links across its boundary. Connectivity here does not require mutual influence between every pair, maximality, an autonomous Markov state, or a nervous system. A numerical renaming of H does not create another physical occurrence.

This is a concrete expression of Section 2.1's process domain, not an independent consciousness detector. Whenever such a process is actual, U1 applies. Different subhistories can overlap and nest; selecting all admissible processes does not demand an exclusive partition. Greater integration, temporal depth, differentiation, or self-maintenance then characterizes differences of organization within the domain.

The representation G must itself be grounded in the physical system. A convenient calculated feature may describe an existing process without creating an additional token. In contrast, an actual causal relation remains relevant even when it crosses an observer's preferred anatomical or computational boundary. Applications may investigate a process's current state carriers, dynamical contributors, enabling conditions, and output recipients separately. Those roles do not determine whether some underlying actual local process has experience.

A cell, a multicellular organism, and an extended interaction can therefore each instantiate experience-bearing organization without passing three different ontological tests. An entire organism need not be one indivisible subject for its many realized processes to fall within UCT. Appendix D gives optional procedures for checking finite coarse descriptions; their predictive-closure conditions are not required by the process criterion or by U1.

3. Ontic organization, views, estimates, and subjects

The central formal firewall of v1.1 is:

$$\text{ontic process token} \neq \text{physical view} \neq \text{scientific estimate} \neq \text{subject}.$$

3.1 Full token-relative ontic organization

For an actual process token P , define:

$$\mathbf{D}_t^{\text{ontic}}(P)$$

as the complete currently instantiated physical organization constitutive of that token.

“Complete” is token-relative. It means that every physical distinction or relation constitutive of P ’s own organization is included. It does not mean that every microscopic fact anywhere in the realizing substrate or surrounding universe is part of the same token.

This distinction permits fine and coarse actual process tokens to coexist without making every microscopic fluctuation a mandatory difference in a human-scale macroexperience.

3.2 Physical scientific views

Science does not ordinarily access $\mathbf{D}^{\text{ontic}}(P)$ directly. Instead, it constructs physically anchored views indexed by:

$$v = (t, \tau, g, \mathcal{I}, \mathcal{B}, \dots),$$

where t is time, τ a horizon, g a measurement/representation grain, $\mathcal{I}$ a physically admissible intervention family, and $\mathcal{B}$ a declared boundary representation.

Write:

$$D_v(P) = \Pi_v[\mathbf{D}^{\text{ontic}}(P)].$$

Different views need not form one total hierarchy. Some are refinements or coarse projections of others; some are incomparable.

Where two routes claim to construct the same target physical view, local consistency requires approximate or exact commutation:

$$R_{v \rightarrow z} \cong R_{w \rightarrow z} \circ R_{v \rightarrow w}.$$

3.3 Scientific estimates

Actual data yield:

$$\widehat{D}_v(P \mid \mathcal{E}, \mathcal{M}),$$

an estimate conditioned on evidence $\mathcal{E}$ and model class $\mathcal{M}$.

If evidence underdetermines the physical organization, the appropriate scientific output is an identified set:

$$\mathfrak{I}(P) = \{D : D \text{ remains compatible with the available evidence}\},$$

rather than an ontological point estimate chosen for convenience.

3.4 Subjecthood

A subject is not defined merely by being a process token, a scientific view, or a macroprocess.

UCT therefore maintains:

experience-bearing process $\neq$ canonically individuated subject.

A universal positive subject selector is not required for this basal theory. Where a study uses subject to mean an experience-bearing process, the process domain supplies its referent. Where it means a human-like owner or conceptual self, the relevant self-organizing and self-representational relations belong to higher-order organization. Neither usage licenses imposing one exclusive subject partition on all actual processes.

3.5 A pointed, port-aware comparison signature

The word *structure* must not conceal what an isomorphism is required to preserve. In the finite controlled models below, the declared signature has sorts for states, boundary inputs, intervention ports, outputs, constituent slots, and time; operation symbols for update, reset, readout, and constituent projection; a distinguished current-state constant; and the declared temporal and incidence relations. A model in that signature can be written:

$$\mathcal{M}_v = (X, x_\bullet, U, J, Y, T, I, o, \mathcal{C}, \preceq).$$

The tuple is a model **in** the signature, not the signature itself. Here $T_u : X \rightarrow X$ is an update at fixed constitutive mechanism, $I_j : X \rightarrow X$ a permitted state reset, and $o : X \rightarrow Y$ a declared output. $\mathcal{C}$ specifies constituent projections and physically identified ports. Current-state pointing is essential: two occurrences of the same transition law at inequivalent current states need not have the same complete organization.

An admissible isomorphism contains bijections f_X, f_U, f_J, f_Y that satisfy

$$f_X(x_\bullet) = x'_\bullet, \quad f_X T_u = T'_{f_U(u)} f_X, \quad f_X I_j = I'_{f_J(j)} f_X, \quad f_Y o = o' f_X,$$

and preserve the declared temporal relations and constituent incidence. A port permutation is allowed only when it is declared as a symmetry of the comparison; an arbitrary mixture of physically distinct ports is not a relabeling of a fixed constituent split. Changing a mechanism, removing an intervention, or reversing a directed temporal relation generally changes the comparison rather than supplying an isomorphism. A probe family chosen by the analyst remains part of a scientific view, whereas a process's actual causal dispositions are ontic.

For the product comparison in Section 6, hold the common boundary convention fixed, use the product state space with its pointed state, and retain the specified component-local update and reset operations. These declarations identify the ordinary independent product used on both sides of C1. They do not impose autonomy, reciprocity, maximality, or a new threshold for experience.

The finite signature supplies an explicit example, not a claim that a finite model captures all constitutive properties of real hardware or a brain. A restricted signature or measured view can omit distinctions. Completeness relative to an actual token is therefore not inferred from success in this finite representation. In biological applications the relevant state carriers, interfaces, and time relations must be justified independently; the same core identity principle is used.

3.6 Structural-type space and invariant topology

Let $\mathcal{C}_{\mathcal{K}}$ be the class/category of admissible $\mathcal{K}$ -structures for a declared comparison. Define the structural-type space:

$$\mathfrak{S}_{\mathcal{K}} = \text{Obj}(\mathcal{C}_{\mathcal{K}}) / \cong.$$

Write:

$$q_{\mathcal{K}}(X) = [X]_{\cong}.$$

For continuity and approximate-structure questions, a topology $\tau_{\mathcal{K}}$, or an isomorphism-invariant pseudometric $\delta_{\mathcal{K}}$, may be fixed on $\mathfrak{S}_{\mathcal{K}}$. It must be declared on structural grounds rather than chosen after observing a desired phenomenal outcome.

Two choices must be separated. First, quotienting by structural isomorphism removes representational relabelings that should not count as new structural types. Second, the topology or pseudometric determines which *different* structural types count as nearby. C1 supplies neither a unique topology nor a unique metric. Consequently, Theorem U2 transfers continuity **within a predeclared structural comparison geometry**; it does not by itself prove that an evolutionary or developmental trajectory is continuous in that geometry.

The use of equivalence classes or quotient spaces is not claimed as new mathematics or as new to consciousness science. Kleiner (2020a) uses automorphism-equivalence classes to characterize intersubjectively well-defined aspects of experience; Kleiner and Ludwig (2024) develop a general framework for mathematical structures of conscious experience, including topological examples; and Oizumi, Lim, and Kanai (2026) use symmetry orbits and a quotient space in a principal-bundle model of perceptual qualia. The present quotient serves a different, narrower role: it identifies **isomorphism classes of the declared physical/experiential structural models** so that the two descriptions in C1 can be compared in one type space. It is not a claim that the present quotient geometry uniquely characterizes qualia.

This quotient construction does not certify that any finite representation is ontically complete. It supplies the common comparison space required by Theorem U2: physical and experiential organizations named by C1 occupy the same structural type rather than two independently topologized spaces.

4. One consciousness-specific core commitment

The v1.1 list A1–A6 is not treated here as an independent axiom basis. P3 and P6 have moved to process ontology; Universal Experience, Structural Continuity, and Selfhood Non-Prerequisite are derived below. The sole consciousness-specific core axiom is the strong tokenwise identity already intended by v1.1.

C1 — Structural-Experiential Identity

For every actual valid process token P , there exists an experiential organization in the same declared structural signature and an isomorphism:

$$\forall P \in \mathcal{P}, \quad \exists \Phi_{\mathcal{K}}(P), \quad h_P : \mathbf{D}_{\mathcal{K}}^{ntic}(P) \xrightarrow{h_P} \Phi_{\mathcal{K}}(P).$$

The two expressions describe the same token-relative organization in physical description and intrinsic experiential presentation. The isomorphism is not an empirically fitted correlation, an extra causal force, a report-generating mechanism, a scalar consciousness score, or a subject-selection rule.

The explicit existential form prevents a notational ambiguity: the theory is not first assuming an already-defined $\Phi(P)$ and then using isomorphism to prove that experience exists. C1 itself is the metaphysical identity commitment from which the existence theorem follows.

This reformulation does not eliminate the substantive universal-experience commitment; it absorbs that commitment into the stronger universal tokenwise identity. A smaller independent axiom list is therefore a reduction in logical redundancy, not an empirical proof from nonphenomenal premises.

Lemma C1-OI — Pairwise type equivalence

Composing tokenwise isomorphisms gives:

$$\boxed{\Phi(P) \cong \Phi(Q) \iff \mathbf{D}^{ontic}(P) \cong \mathbf{D}^{ontic}(Q).}$$

This pairwise equivalence is weaker than C1 written alone: an arbitrary permutation of type labels could preserve a biconditional without preserving each token's internal organization.

Lemma C1-W — One-way phenomenal completeness

The one-way consequence is:

$$\boxed{\mathbf{D}(P) \cong \mathbf{D}(Q) \implies \Phi(P) \cong \Phi(Q).}$$

C1-W suffices for No Hidden Quale, Probe Invariance, No Future Contamination, and Causally Screened History.

Theorem U1 — Universal Nonempty Experience

For every actual valid process token:

$$\boxed{P \in \mathcal{P} \Rightarrow \Phi(P) \neq \emptyset.}$$

Proof. By the process-token criterion, an actual token contains at least one physically instantiated event/state in the constitutive carrier sort. Hence the corresponding ontic $\mathcal{K}$ -structure is nonempty in that distinguished sort. C1 supplies a sort-preserving isomorphism to $\Phi_{\mathcal{K}}(P)$. A bijection from a nonempty carrier cannot have an empty codomain carrier. Therefore $\Phi(P) \neq \emptyset$. $\square$

The distinguished-carrier qualification matters in a many-sorted signature: auxiliary sorts may legitimately be empty, whereas an actual token cannot have an empty constitutive event/state carrier.

U1 replaces v1.1 A1 as an independent axiom. Universal experience remains part of UCT's substantive content because C1 is itself universal over actual tokens.

Theorem U2 — Structural Continuity

Let Λ be a parameter space and P_{λ} an explicitly comparable family of actual process tokens. Define:

$$d(\lambda) = q_{\mathcal{K}}(\mathbf{D}_{\mathcal{K}}^{ontic}(P_{\lambda})), \quad \phi(\lambda) = q_{\mathcal{K}}(\Phi_{\mathcal{K}}(P_{\lambda})).$$

By C1:

$$d(\lambda) = \phi(\lambda) \quad \forall \lambda \in \Lambda.$$

Therefore, for any predeclared topology $\tau_{\mathcal{K}}$ on the common structural-type space:

$$d \text{ continuous} \iff \phi \text{ continuous.}$$

For any predeclared isomorphism-invariant pseudometric:

$$\delta_{\mathcal{K}}(d(\lambda), d(\mu)) = \delta_{\mathcal{K}}(\phi(\lambda), \phi(\mu)).$$

Proof. C1 places the physical and experiential organizations of each token in the same isomorphism class. Hence the quotient-valued maps d and ϕ are pointwise identical. Identical functions have the same topological and metric regularity properties. $\square$

U2 concerns structural continuity. It does not imply a universal scalar amount of consciousness, integer subject-count continuity, reportability continuity, conceptual-self continuity, or continuity of externally chosen human qualitative labels.

U2 is also a **transfer theorem**, not a proof that nature supplies a continuous path. Establishing that a biological, developmental, artificial, or replacement trajectory $\lambda \mapsto P_{\lambda}$ is continuous in the chosen $(\mathfrak{S}_{\mathcal{K}}, \tau_{\mathcal{K}})$ remains a physical/modeling premise. Choosing a topology after seeing a desired experiential conclusion would violate the same no-post-hoc discipline used elsewhere in the paper.

Corollary U3 — Selfhood Non-Prerequisite

Suppose there exists an actual token P_0 lacking human-like conceptual selfhood:

$$P_0 \in \mathcal{P}, \quad \neg \text{Self}_I(P_0).$$

By U1:

$$\Phi(P_0) \neq \emptyset.$$

Therefore human-like conceptual selfhood is not a necessary condition for basal experience.

The logical form is explicitly witness-dependent. U3 requires at least one actual token for which the chosen operational/structural definition of human-like conceptual selfhood is absent. The existence and correct classification of such a witness are empirical/semantic premises; they are not supplied by the theorem itself.

The positive mechanisms by which body models, agency, autobiographical memory, metacognition, and a conceptual “I” form remain higher-order application/bridge questions.

Migration table

v1.1 item	v1.2 status
A1 Universal Experience	Theorem U1
A2 Scoped Structural Continuity	Theorem U2 on $\mathfrak{S}_{\mathcal{K}}$
A3 Persistence Under Embedding	Process Ontology Rule P3
A4 Selfhood Non-Prerequisite	Corollary U3
A5-SI Experience-Structure Identity	Core Axiom C1

v1.1 item	v1.2 status
A5-OI / A5-W	Lemmas C1-OI / C1-W
A6 Token / Type / Lineage	Process Ontology Definition P6

Proposition U0 — Foundational Compression

Fix the process ontology of Section 2, the common structural signature $\mathcal{K}$, and the structural-type space of Section 3. Then the v1.1 foundational roles can be reconstructed as follows:

$$\begin{aligned} C1 + \text{nonempty actual-token carrier} &\Rightarrow U1, \\ C1 + \text{predeclared structural geometry} + \text{physical path continuity} &\Rightarrow U2, \\ U1 + \text{non-self actual-token witness} &\Rightarrow U3. \end{aligned}$$

P3 and P6 are not additional consciousness-existence principles; they specify which physical process token is being discussed and how token identity/lineage is tracked.

Proof. The first line is Theorem U1. The second is Theorem U2. The third is Corollary U3. P3 and P6 enter only through the domain and identity conditions for the relevant tokens. $\square$

The proposition concerns **logical dependency**, not empirical truth. Any finite list of assumptions could be trivially conjoined into one sentence; such syntactic compression would have no theoretical significance. The present minimization is substantive only to the extent that U1, U2, and U3 are actually derivable from C1 plus independently typed ontological/structural premises rather than being hidden as separate consciousness axioms.

5. Formal consequences

5.1 Organization-Gate Impossibility

Define an experience-existence predicate:

$$E(P) = \mathbf{1}[\Phi(P) \neq \emptyset].$$

By Theorem U1:

$$E(P) = 1 \quad \forall P \in \mathcal{P}.$$

Let $F(P)$ be any organization descriptor and S a threshold/target region. If:

$$\exists Q \in \mathcal{P} : F(Q) \notin S,$$

then:

$$\mathbf{1}[F(P) \in S] \neq E(P).$$

Hence, within UCT, no nonuniversal threshold of:

- integration;
- recurrence;
- global availability;
- explicit memory;
- higher-order representation;

- predictive modeling;
- autonomy;
- entropy;
- generic biological complexity;

can equal the universal experience-existence predicate.

This theorem does not make those mechanisms irrelevant. It relocates them from experience existence to experience organization.

5.2 No Hidden Quale

From C1-W:

$$\mathbf{D}^{ontic}(P) \cong \mathbf{D}^{ontic}(Q) \Rightarrow \Phi(P) \cong \Phi(Q).$$

UCT therefore rejects an additional hidden phenomenal degree of freedom that varies independently while complete token-relative physical organization is fixed.

Within the intended structuralist interpretation, words such as “red” and “pain” label experiential organizations rather than additional freely varying quale atoms. C1-W alone does not determine the physical organization corresponding to either label or establish the complete structuralist interpretation.

5.3 Probe Invariance

Let $\mathcal{J}^{exp}$ denote an experimenter-selected probe family. Merely changing this choice, without changing the complete constitutive physical organization, cannot alter phenomenology under C1-W:

$$\begin{aligned} \mathcal{J}_1^{exp} &\neq \mathcal{J}_2^{exp}, \\ \mathbf{D}_1^{ontic}(P) \cong \mathbf{D}_2^{ontic}(P) &\implies \Phi_1(P) \cong \Phi_2(P). \end{aligned}$$

The completeness qualifier matters. Holding only one causal graph or one response descriptor fixed is insufficient if other constitutive state distinctions change. A probe that physically intervenes on the process can change its experience by changing its organization; the invariant claim concerns a merely external change of description or choice.

5.4 No Future Contamination

If:

$$\mathbf{D}_t^{ontic}(P) \cong \mathbf{D}_t^{ontic}(Q),$$

then a future external divergence that is not physically instantiated at time t cannot imply a present phenomenal difference:

$$\boxed{\Phi_t(P) \cong \Phi_t(Q)}.$$

Present causal dispositions are part of the present organization; actual future facts are not.

5.5 Causally Screened History / No Ghost History

Suppose:

$$History(P) \neq History(Q)$$

but all current constitutive physical consequences of those histories have converged:

$$\mathbf{D}_t^{ontic}(P) \cong \mathbf{D}_t^{ontic}(Q).$$

Then:

$$\boxed{\Phi_t(P) \cong \Phi_t(Q).}$$

P6 still permits the tokens to remain numerically distinct because lineage can differ even when current structural type is the same.

5.6 External reward-label non-identity

Changing an external reward sign, semantic label, or experimenter convention without changing intrinsic constitutive organization cannot by itself alter phenomenology under C1-W.

Therefore:

$$\boxed{\text{reward sign} \neq \text{phenomenal valence sign}.}$$

6. Factorization, macroprocesses, and combination

6.1 Independent products and comparison objects

Fix structural signatures and a constituent split before comparing a whole with its parts. The physical product $X_1 \otimes \cdots \otimes X_n$ represents independent juxtaposition relative to that split and a declared boundary convention. In a transition model, component updates must not depend on the other component states once the declared boundary is fixed. A shared drive can create statistical correlations without creating the internal causal dependence at issue. Product factorization is consequently not synonymous with unconditional statistical independence.

For this product comparison, constituents occupy nonoverlapping slots of the specified physical split. A product of overlapping actual tokens would require a separately justified comparison construction; general gluing of overlapping processes is not established here. An abstract product is initially a **comparison object**. It need not itself be an actual causally continuous token; U1 is not automatically applied to it. Unrestricted mixing of physically distinct intervention ports must not be mistaken for a mere relabeling that preserves a particular constituent split. The phenomenal product must also be specified as the intended independent juxtaposition, not retrospectively defined to make a desired conclusion true.

6.2 Why C1-OI alone is insufficient

C1-OI constrains equivalence of structural types. It does not state how products are transported. The distinction is visible in a finite-set countermodel. Take isomorphism classes indexed by positive cardinality and the bijection

$$f(6) = 7, \quad f(7) = 6, \quad f(n) = n \quad (n \neq 6, 7).$$

Let physical constituent types have sizes 2 and 3, and a whole type have size 7. Map phenomenal types using f . The equivalence biconditional holds because f is injective. Nevertheless,

$$7 \neq 2 \cdot 3, \quad f(7) = 6 = f(2) \cdot f(3).$$

Thus a nonproduct physical type can correspond to the product of the constituent phenomenal types while the bare C1-OI equivalence formula is satisfied. This is an abstract countermodel to an implication of the formula, not a claim that unstructured finite sets are adequate physical or phenomenal models. It pinpoints the missing algebraic commitment. Additional physical restrictions may exclude it, but those restrictions must be stated.

6.3 Non-summative combination from the intended identity

Use the strong tokenwise identity C1 in a common comparison category with the finite products needed for the argument. Independent juxtaposition on each side is represented by this same standard product. Ordinary products preserve isomorphisms: if each $h_i : X_i \cong Y_i$, then

$$\bigotimes_i h_i : \bigotimes_i X_i \cong \bigotimes_i Y_i.$$

This is the relevant mathematical consequence of expressing the same constituent organizations in two descriptions. It is not another physical mechanism of experiential combination. The use of a common signature and product must be declared; the weak type-equivalence formula alone cannot supply it.

Theorem 6.3 (non-summative combination in the declared product comparison).

Let M be an actual whole process and P_i its actual constituents in the nonoverlapping slots of the specified comparison. If

$$\mathbf{D}^{ontic}(M) \not\cong \bigotimes_i \mathbf{D}^{ontic}(P_i),$$

then C1 implies

$$\Phi(M) \not\cong \bigotimes_i \Phi(P_i).$$

Proof. Let h_M and h_i be the tokenwise isomorphisms supplied by C1. If $\Phi(M) \cong \bigotimes_i \Phi(P_i)$, compose that isomorphism with h_M and the inverse of the product of the h_i . This yields

$$\mathbf{D}^{ontic}(M) \cong \Phi(M) \cong \bigotimes_i \Phi(P_i) \cong \bigotimes_i \mathbf{D}^{ontic}(P_i),$$

contradicting the physical premise. The abstract comparison product need not be an actual token, so no extension of U1 to imaginary processes is used. $\square$

The result says that actual additional constitutive relations are not erased in the experiential description. It does not add together subjective substances. Its nonproduct conclusion is relative to the specified constituent product, not every possible decomposition. Continuing constituent processes remain experience-bearing by U1 and P3. Neither the theorem nor the theory requires a unique numerical subject count before admitting the whole's experiential organization.

If identity is deliberately weakened back to the pairwise C1-OI formula alone, the countermodel in Section 6.2 still applies and this proof is unavailable. The repair therefore

makes the intended premise explicit rather than silently inferring a stronger premise from a weaker formula.

6.4 Macro invariance with micro change

Suppose actual macroprocess tokens obey

$$\mathbf{D}^{ontic}(M_A) \cong \mathbf{D}^{ontic}(M_B),$$

while an independently identified pair of lower-scale actual tokens obeys

$$\mathbf{D}^{ontic}(p^A) \not\cong \mathbf{D}^{ontic}(p^B).$$

C1-W suffices for $\Phi(M_A) \cong \Phi(M_B)$. The converse component of C1-OI additionally yields $\Phi(p^A) \not\cong \Phi(p^B)$. Only the stated pairwise consequences are needed for these claims; the product construction is not involved.

The result is conditional: it does not show that any specified material replacement actually preserves complete macro organization. Nor can one establish it merely by observing the same coarse statistic of the same token. The macro token must have an independently supported realization and boundary; lower-scale physical differences must not be constitutive differences of that macro token. Appendix B demonstrates the corresponding distinction between a fixed whole-process token and an invariant implemented sub-process in an idealized finite model. It is not a general biological replacement theorem.

7. Subject selection under symmetry: an orbit obstruction

UCT distinguishes experience, structural unity, macroprocess tokenhood, subjecthood, and self-modeling. The result below concerns a proposed selection rule. It neither derives subjecthood from U1 nor treats every experience-bearing process as one canonical subject.

7.1 Equivariant selection is a union of candidate orbits

Let W be a physical structure and Γ a group of its structure-preserving automorphisms. Let $\mathcal{C}(W)$ be a candidate family closed under this group. A deterministic selector returns $S(W) \subseteq \mathcal{C}(W)$ and is equivariant:

$$S(gW) = gS(W).$$

Since $gW = W$ for $g \in \Gamma$, the selected family is invariant under Γ .

Orbit lemma. If a candidate P is selected, its entire orbit ΓP is selected.

Proof. For each $g \in \Gamma$, invariance gives $gP \in gS(W) = S(W)$. Thus a selected family is a union of full orbits. $\square$

Corollary 7.1 (restricted exclusive-selection obstruction). Suppose the admissible candidates are one orbit O containing an overlapping pair. No deterministic equivariant selector can return a nonempty, pairwise disjoint subfamily of O .

Proof. A nonempty selection contains one candidate and hence all of O . It then contains the overlapping pair, contrary to pairwise disjointness. $\square$

This corollary requires the stated restriction of the candidate family and a nonempty-output requirement. These are selection requirements, not consequences of the process ontology plus C1/U1/U2.

7.2 The three-part example and its counterlimits

For a reflection-symmetric world with parts a, b, c , let the reflection exchange a and c , and let the candidate orbit be

$$O = \{\{a, b\}, \{b, c\}\}.$$

The two invariant selected subfamilies are the empty family and the whole orbit. The first violates nonemptiness; the second violates exclusivity. An unbiased random rule could choose either member with equal probability, but it would not determine one unique subject from W ; a realized random seed would add selection data.

The theorem does **not** rule out the invariant partitions

$$\{\{a, b, c\}\}, \quad \{\{a\}, \{b\}, \{c\}\}.$$

If a positive subject theory admits one of these, it needs its own physical justification. Neither is excluded by the orbit lemma. The correct conclusion is a scoped obstruction to selecting nonempty exclusive candidates from an overlapping symmetry orbit, not a general theorem that intrinsic structure cannot determine any unique partition. A universal positive subject criterion is not imposed by UCT. Any additional exclusive selector must justify its own scope and purpose.

8. No first conscious ancestor: from physical process to conceptual self

This section is explanatory rather than foundational. Its central UCT consequence is:

within the UCT domain, there is no first conscious ancestor.

UCT does not describe evolution as a transition from nonexperience to experience. It describes transformations among already experiential actual processes. The simplest actual physical interaction processes have nonempty experiential organization by U1; chemical systems, cells, multicellular systems, nervous systems, animals, and humans instantiate progressively different organizations under the same C1 principle.

This claim concerns **actual processes**, not inert labels such as “particle,” “atom,” or “organism” taken in abstraction. An elementary-particle interaction, an atomic process, a cell, or a brain is in the UCT domain only insofar as the relevant actual process satisfies the token criteria of Section 2.

8.1 Corollary E1 — No First Conscious Ancestor

Define the experience-existence predicate:

$$E(P) = \mathbf{1}[\Phi(P) \neq \emptyset].$$

By U1:

$$E(P) = 1 \quad \forall P \in \mathcal{P}.$$

Let

$$\gamma : \Lambda \rightarrow \mathcal{P}$$

be a family all of whose members are **actually instantiated valid process tokens in the scenario under evaluation**. Then:

$$E(\gamma(\lambda)) = 1 \quad \forall \lambda \in \Lambda.$$

For the All-Ancestors-Survive thought experiment below, simultaneous actualization of the comparison family is part of the stipulation: one imagines a physically admissible scenario in which representatives of the intermediate organizations are all instantiated. U1 is applied only to those stipulated actual tokens, not to a merely drawn state-space point or an uninstantiated mathematical possibility.

Therefore no λ_* inside the UCT domain can mark a transition of the form:

$$E(\gamma(\lambda < \lambda_*)) = 0, \quad E(\gamma(\lambda > \lambda_*)) = 1.$$

Proof. Every $\gamma(\lambda)$ is, by construction, an actual valid process token. U1 applies to every such token, so $\Phi(\gamma(\lambda)) \neq \emptyset$ for every λ . Hence $E(\gamma(\lambda)) = 1$ throughout. $\square$

This result does **not** say that every stage has the same experience, the same richness, the same unity, or the same cognitive capacities. It states only that the experience-existence predicate does not switch from zero to one inside the declared UCT domain.

Corollary E1 is not claimed as historically novel merely because it rejects a first conscious biological threshold. Panexperientialist and panpsychist traditions have long treated experience or mentality as fundamental and widespread. The contribution claimed here is the explicit derivation of this no-ignition result inside UCT's process-token/structural-identity dependency architecture, not the general idea that experience extends below animals or brains.

There may be genuine organizational thresholds for:

- recurrent processing;
- long-range access;
- memory;
- predictive control;
- self-modeling;
- metacognition;
- language;
- conceptual autobiographical selfhood.

Those are transitions **within experiential organization**, not transitions from no experience to experience.

8.2 The All-Ancestors-Survive Thought Experiment

Imagine that representatives of every sufficiently fine intermediate form in the physical history from simple actual interactions through stable chemistry, early cells, multicellular organization, nervous systems, animals, and humans were simultaneously instantiated and available for comparison.

"Ancestor" is used schematically in the name of this thought experiment. Elementary physical and prebiotic chemical stages are not biological ancestors in the strict Darwinian sense. The argument joins a physical/chemical prehistory to a biological evolutionary lineage; it does not claim that an electron or molecule is literally an organismal ancestor.

Represent the family schematically as:

$$P_0, P_1, \dots, P_n,$$

or, in a finer idealization,

$$\gamma : \Lambda \rightarrow \mathcal{P}.$$

Suppose neighboring members can be chosen so that their relevant physical organizations differ only slightly in a predeclared structural geometry:

$$\delta_{\mathcal{K}}(d(\lambda), d(\lambda + \Delta\lambda)) \rightarrow 0 \quad \text{as } \Delta\lambda \rightarrow 0.$$

UCT does not search this family for an index i^* satisfying:

$$E(P_{i^*-1}) = 0, \quad E(P_{i^*}) = 1.$$

Corollary E1 has already ruled out such an onset within the UCT domain. The scientific question is instead:

Which physically instantiated distinctions and relations are added, removed, re-organized, stabilized, or made available as the process family changes?

If the structural-type path

$$d(\lambda) = [\mathbf{D}_{\mathcal{K}}^{ontic}(\gamma(\lambda))]_{\cong}$$

is continuous in the predeclared invariant topology on $\mathfrak{S}_{\mathcal{K}}$, U2 gives:

$$\phi(\lambda) = [\Phi_{\mathcal{K}}(\gamma(\lambda))]_{\cong} = d(\lambda),$$

and therefore:

$$\boxed{d \text{ continuous} \implies \phi \text{ continuous.}}$$

The intended picture is:

$$\boxed{\text{nonempty experience throughout} + \text{changing experiential organization.}}$$

This is the formal expression of the core evolutionary intuition motivating UCT.

Corollary E2 — Fine-Grained Structural Chains

The user's intended continuity need not be an exactly differentiable or even globally continuous biological path. Fix an isomorphism-invariant pseudometric $\delta_{\mathcal{K}}$. Suppose a finite actual-token chain

$$P_0, P_1, \dots, P_n$$

satisfies:

$$\delta_{\mathcal{K}}(d(P_i), d(P_{i+1})) \leq \varepsilon \quad (i = 0, \dots, n-1).$$

Because C1 identifies the physical and experiential descriptions with the same points of the structural-type space:

$$\delta_{\mathcal{K}}(\phi(P_i), \phi(P_{i+1})) = \delta_{\mathcal{K}}(d(P_i), d(P_{i+1})) \leq \varepsilon.$$

Hence every ε -fine physical structural chain is an equally ε -fine experiential structural chain.

If the comparison family can be refined so that such chains exist for arbitrarily small $\varepsilon > 0$, the experiential comparison can be refined to the same degree. This is weaker than assuming one globally smooth evolutionary trajectory and matches the intended claim that the physical/experiential development can be **gradual without requiring mathematically perfect continuity**.

E2 does not imply that complexity or one scalar “amount of consciousness” increases monotonically. A chain can add, delete, prune, bifurcate, specialize, or reorganize relations while remaining finely graded.

Explanatory pressure on ignition theories

A competing theory may instead posit an experience-onset condition G . In the all-ancestors construction, if two neighboring organizations can be made arbitrarily similar while the competitor assigns:

$$E(P_i) = 0, \quad E(P_{i+1}) = 1,$$

then that theory owes an account of which physically real difference grounds the zero-to-nonzero change and why that difference has **experience-existence** significance rather than merely organizational significance.

This is not a formal contradiction in the competing theory. It is an explanatory burden. UCT does not incur that additional ignition cost because C1/U1 already treat every actual valid process as experiential.

8.3 Why Evolutionary Continuity Alone Does Not Prove UCT

The preceding result is an **internal UCT consequence**, not a derivation of C1 from Darwinian continuity alone.

A theorist who rejects C1 can accept arbitrarily fine physical continuity while assigning a separate onset rule to experience. For example, even if one grants a scalar proxy $c(\lambda)$ solely to construct a countermodel and demands perfect smoothness, one may define:

$$c(\lambda) = \begin{cases} 0, & \lambda \leq \lambda_*, \\ \exp\left[-\frac{1}{(\lambda - \lambda_*)^2}\right], & \lambda > \lambda_*. \end{cases}$$

This C^∞ function remains exactly zero before λ_* and becomes positive smoothly afterward. Hence:

$$\boxed{\text{physical continuity alone} \not\Rightarrow \text{universal nonempty experience.}}$$

The scalar c is **not** a UCT consciousness measure, and this construction is **not** UCT’s positive model. Its sole purpose is to block an invalid external inference from “evolution is continuous” directly to “all stages are experiential.”

There is, however, a precise theorem corresponding to the intuition that a genuinely continuous **existence predicate** cannot switch on halfway through a connected ancestor spectrum.

Conditional Lemma E3 — Connected Existence Constancy

Let Λ be connected and let:

$$E_\gamma : \Lambda \rightarrow \{0, 1\}$$

be the experience-existence predicate along a comparison family, where $\{0, 1\}$ carries the discrete topology. If E_γ itself is continuous, then E_γ is constant. Therefore, if one endpoint is a conscious human with:

$$E_\gamma(\lambda_h) = 1,$$

then:

$$E_\gamma(\lambda) = 1 \quad \forall \lambda \in \Lambda.$$

Proof. The continuous image of a connected space is connected. The only nonempty connected subsets of the discrete space $\{0, 1\}$ are the singletons. Hence the image of E_γ is either $\{0\}$ or $\{1\}$. The human endpoint selects $\{1\}$. $\square$

This lemma explains the force of the continuity intuition, but it also identifies its additional premise: continuity of **experience existence itself**, not merely continuity of physical organization. UCT does not need to add that premise independently because U1 already fixes $E = 1$ throughout its actual-process domain.

The correct UCT dependency is:

$$C1 \Rightarrow U1 \Rightarrow \text{no experience-existence ignition within the UCT domain.}$$

U2 then adds structural continuity whenever the physical path is continuous in the predeclared comparison geometry.

Thus the all-ancestors thought experiment has two legitimate roles:

1. **within UCT:** it illustrates a continuously or finely transforming spectrum of already nonempty experiential organization;
2. **against threshold theories:** it exposes the extra explanatory burden of locating and justifying an experience-existence onset.

It is not used as an independent proof of C1.

The distinction is therefore exact: **E2** captures graduality without demanding a mathematically perfect continuum; **E3** shows that if a rival additionally assumes continuity of the binary experience-existence predicate itself, then the user's intuition becomes a formal constancy theorem. Neither result replaces C1 in UCT.

8.4 Elementary physical processes

At the most elementary level considered by UCT, an actual elementary physical process token possesses whatever state, causal, temporal, dispositional, or relational organization is physically instantiated. The present paper does not commit to a final ontology of particles or fields; a particle interaction, worldline segment, field process, or quantum process would require the corresponding physical completion of the token formalism.

UCT does not require:

- a nervous system;
- a sensor;
- a memory;
- a self-model;
- a report channel.

Under U1, the experience-existence predicate is already nonzero. C1 identifies this process's experiential organization with its actual organization. The pairwise C1-OI formula is a consequence of that stronger statement.

UCT therefore refuses to project human terms such as “seeing,” “wanting,” or “suffering” onto elementary interactions. The primitive claim is structural, not anthropomorphic.

8.5 From stable chemistry to self-maintaining organization

The transition from nonliving chemistry to the first cells is an origin-of-life problem, not a solved Darwinian lineage. UCT does not claim to reconstruct the historical path.

The theoretical point is narrower. If physical processes develop persistent boundaries, recurrent chemical organization, and endogenous regulation, then new relations appear inside the process organization:

$$\text{internal state} \leftrightarrow \text{boundary} \leftrightarrow \text{environmental perturbation} \leftrightarrow \text{regulatory response}.$$

These relations are not the origin of experience under U1. They are transformations of experiential organization.

8.6 Single-cell and non-neural learning: temporal depth

A decisive organizational transition occurs when past events leave present physical traces that modify current response:

$$\text{History} \rightarrow m_t \rightarrow \text{Response}_t.$$

This is the minimal form of temporal depth required by UCT’s memory analysis.

The framework does not require memory to create experience. Memory changes current organization.

Non-neural learning illustrates the physical possibility of history-dependent behavior without a brain. Habituation has been experimentally demonstrated in the slime mould *Physarum polycephalum* (Boisseau, Vogel, & Dussutour, 2016). Such evidence does not establish phenomenology in the organism; it demonstrates that learning-like temporal organization need not await centralized nervous systems.

If all physical consequences of a history are erased, No Ghost History predicts no remaining current phenomenal-type difference attributable solely to the past.

8.7 Multicellularity: from juxtaposition to organism-scale relations

Consider independent cells:

$$C_1, \dots, C_n.$$

If their joint organization remains a product, there is no new irreducible cross-cell relation merely because the cells are spatially adjacent.

If, however, real chemical, electrical, mechanical, developmental, or other causal relations generate an actual multicellular process M with:

$$\mathbf{D}(M) \not\cong \bigotimes_i \mathbf{D}(C_i),$$

then the structure-for-structure identity C1 assigns M an experiential organization that is nonproduct relative to that constituent comparison, by Theorem 6.3. The multicellular case uses the same identity principle as the cellular case; no new experiential substance or threshold is introduced.

The significance of multicellularity in UCT is therefore not cell count. It is the formation of new actual macroprocess organization.

No privileged signaling medium is required. Mechanical, chemical, electrical, and neural relations can all contribute when they are physically constitutive.

8.8 Nervous systems: routing and specialization, not an experience switch

The evolution of neurons adds high-speed, long-range, selectively routed, and plastic coordination.

Under UCT:

neuron $\neq$ experience ignition.

Neurons transform organization. They can increase spatial discrimination, routing, temporal coordination, learning capacity, and context sensitivity.

Associative learning in the box jellyfish *Tripedalia cystophora*, a cnidarian without a vertebrate-style centralized brain, provides an empirical illustration that nontrivial learning need not wait for vertebrate-like centralization (Bielecki et al., 2023).

This finding is not evidence for UCT's U1. It illustrates the danger of equating a particular level of centralized neural architecture with the first appearance of complex learning.

8.9 Damage regulation, nociception, and affect

The pain lineage is particularly useful because it exposes the mistake of projecting a fully human quale backward.

Comparative biology indicates that damage detection and adaptive cellular stress responses predate nervous systems, while specialized nociceptors, sensitization, central aversive states, and human pain involve progressively richer mechanisms (Walters, 2025).

A schematic structural trajectory proceeds through:

1. cellular damage regulation;
2. multicellular injury coordination;
3. specialized nociception and sensitization;
4. aversive learning and multidimensional affective pain;
5. human suffering.

UCT does not identify the first term with human pain.

It instead asks which physical relations constituting later pain are already present in simpler systems.

Human pain also illustrates why one scalar valence variable is inadequate. Clinical and conceptual literatures distinguish sensory-discriminative from affective-motivational dimensions of pain, and pain-asymbolia cases have been interpreted as evidence for partial dissociation between them (Auvray, Myin, & Spence, 2010). That interpretation is not uncontested: recent philosophical reviews emphasize that asymbolia may reflect strongly reduced rather than literally absent negative affect, and that the taxonomy of the residual state remains debated (Coninx, 2024). The conservative UCT conclusion is therefore:

Nociceptive discrimination, unpleasantness, and suffering should not be collapsed into one primitive scalar variable.

Under C1, unpleasantness belongs to the intrinsic organization of the relevant actual process, just as any cellular quality does. Identifying which concrete human mechanisms instantiate a specified pain experience is an application-level biological question; it does not require another foundational account of why human experience exists.

One explanatory principle across experiential qualities

Let P_c be a cellular process and P_h a human process. UCT applies the same relation $\Phi_{\mathcal{K}}(P) \cong \mathbf{D}_{\mathcal{K}}^{ontic}(P)$ in each appropriately declared comparison. The organizations differ in their realized distinctions and relations; the explanatory principle does not change. Human red, pain, and bodily ownership can occupy richly differentiated positions within a process's experiential organization without requiring that a cell possess those same human discriminations or linguistic labels.

In a continuously comparable transformation family, U2 supplies experiential continuity and C1 supplies structural identity. Added, removed, or reorganized causal relations correspond to changes in the intrinsic organization itself. The theory need not posit a point where an otherwise nonexperiential organization suddenly acquires a new phenomenal substance. Nor does it follow that evolution must monotonically increase one scalar amount of experience.

At this explanatory level, asking for a second mechanism connecting a fully specified actual organization to its experience would reopen what C1 fixes as the identity principle. A separate question is which actual organization realizes a particular familiar human experience. Investigating that empirical instance can refine an application without constituting a missing axiom of the general theory.

8.10 Association, relational abstraction, and reasoning

Cognitive evolution can likewise be represented as transformations of organization rather than the sudden arrival of a new phenomenal substance.

A simplified comparative sequence runs from state-dependent response through habituation and sensitization, association, relational abstraction, relations over relations, and symbolic or compositional reasoning.

The sequence is not a literal phylogenetic ladder.

Honeybees can learn and transfer same/different relational rules to novel stimuli (Giurfa et al., 2001), supporting the narrower claim that transferable relational discrimination need not require a human-sized brain.

Cuttlefish have shown what-where-when behavioral organization meeting standard criteria for episodic-like memory (Jozet-Alves, Bertin, & Clayton, 2013), and later work demonstrated substantial delayed control in a delay-of-gratification task (Schnell et al., 2021).

These examples support the narrower conclusion that component capacities relevant to human cognition occur in distant lineages and can be realized by substantially different nervous-system architectures. They do not establish a single linear ancestry of cognition, nor do they demonstrate that nonhuman animals execute human formal logic or possess phenomenology identical to human cognition.

8.11 Physical witness and semantic inflation

A danger in comparative cognition is to redescribe simple behavior with sophisticated human concepts.

UCT therefore introduces a Physical Witness Criterion:

A claimed higher-order relation belongs to the actual constitutive organization only if the physical process itself distinguishes that relational configuration under controlled, physically admissible perturbation.

For example, "same/different" should not be attributed merely because an analyst can describe stimuli relationally. Transfer to novel physical stimuli provides stronger evidence that the system is sensitive to an invariant relation, though behavioral transfer alone still need not reveal the complete ontic mechanism.

The criterion blocks:

observer's description $\Rightarrow$ invented phenomenal dimension.

8.12 Body-centered organization

With increasingly coordinated sensorimotor organization, a system can physically encode relations involving:

- internal/external boundary;
- body position;
- sensor position;
- action state;
- consequences of self-generated action;
- persistent internal variables.

Write this only as a projection of already existing process organization:

$$\mathcal{S}(M) = \Pi_{sys}[\mathbf{D}^{ontic}(M)].$$

This is system-relative organization, not a new observer inside the system.

It can support increasingly rich body-centered and agency-related experience without being the origin of experience itself.

8.13 Self-model

A self-model is a high-level physical organization in which some internal states track, predict, compress, or regulate aspects of the system's own organization across time.

Possible ingredients include:

- body state;
- action consequences;
- memory;
- prediction;
- social information;
- control;
- metacognitive access.

UCT does not define the self-model as the subject, and it does not make self-modeling necessary for basal experience.

8.14 The conceptual "I"

Human conceptual selfhood adds language, autobiographical structure, explicit temporal narration, symbolic categorization, social cognition, metacognition, and culturally transmitted conceptual tools.

A system can then instantiate structures corresponding to propositions such as:

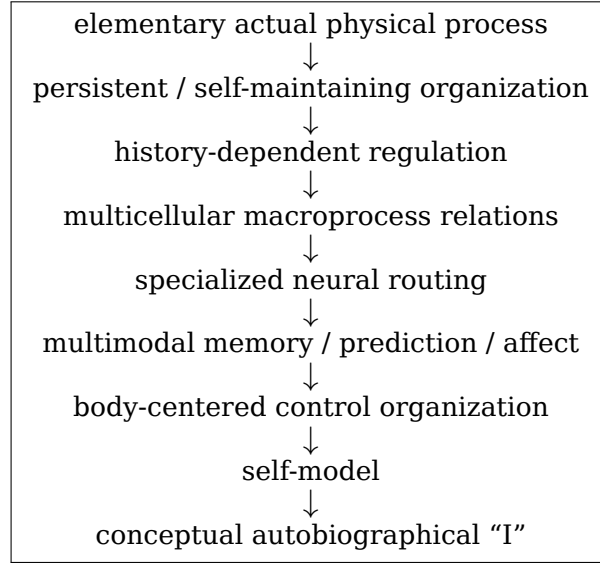
I am the one undergoing this experience.

or:

I remember being the same individual who experienced that event.

Within UCT, this conceptual "I" is a late, high-dimensional organization inside experience, not the entity that creates experience.

The proposed structural sequence is therefore:



The arrows mean structural transformation, not one linear species ancestry.

The comparative cases in this section are therefore evidential anchors for organizational possibilities and intermediate capacities. They are not empirical confirmation of universal experience, structural identity, or cross-species quale identity.

8.15 No universal consciousness ladder

Evolution can:

- add relations;
- delete relations;
- reweight couplings;
- prune structure;
- specialize;
- compress irrelevant details;
- introduce new temporal dependencies;
- bifurcate dynamically;
- lose capacities;
- converge on similar functions by different architectures.

Therefore UCT rejects:

$$\frac{dC}{dt} > 0$$

for a universal scalar amount of consciousness.

The more general representation is:

$$\lambda \mapsto \mathbf{D}_{\lambda}^{ontic}(P_{\lambda}) \mapsto \Phi_{\lambda}(P_{\lambda}).$$

Evolution is a path through a branching space of experiential organization, not a staircase from unconscious matter to conscious humans.

9. Structural transformation versus enrichment

A previous development branch emphasized relational enrichment:

$$D_1 \prec_E D_2.$$

Strict enrichment remains useful when D_2 preserves the distinctions of D_1 while adding new, nonredundant constitutive relations. But biological and artificial change is more general.

Let:

$$T : \mathbf{D}_1 \mapsto \mathbf{D}_2.$$

Possible transformations include current-state change, coupling reweighting, relation addition or deletion, pruning, factorization or refactorization, temporal-depth change, abstraction/compression, dynamic bifurcation, and specialization.

Thus:

$$\boxed{\text{enrichment} \subset \text{structural transformation.}}$$

Under C1-OI:

$$\mathbf{D}_1 \not\cong \mathbf{D}_2 \Rightarrow \Phi_1 \not\cong \Phi_2.$$

This is a direct consequence of the identity commitment, not an independent empirical law.

9.1 An exact edge-addition family with unchanged declared output

Extend Appendix B by controlling two cross-links separately:

$$s^+ = s \oplus u, \quad a^+ = s \oplus (\alpha b), \quad b^+ = s \oplus (\beta a), \quad y = s, \quad (\alpha, \beta) \in \{0, 1\}^2.$$

The fixed comparison contains the whole network $W_{\alpha\beta}$, the implemented s -process, and the boundary-conditioned $a \mid b$ constituent split. The settings are distinct mechanisms, not one continuous U2 interpolation.

For common s_0 and input string, all four settings have the same declared output. Fix the common boundary trajectory s_t , allow all four initial pair states $z_0 = (a_0, b_0)^T$, and apply no further pair resets. Pair differences obey:

$$\delta z_n = A_{\alpha\beta}^n \delta z_0, \quad A_{\alpha\beta} = \begin{pmatrix} 0 & \alpha \\ \beta & 0 \end{pmatrix}.$$

Define retained distinguishability at horizon $n \geq 1$:

$$N_n = |\{z_n : z_0 \in \mathbb{F}_2^2\}| = 2^{\text{rank}(A_{\alpha\beta}^n)}.$$

This is a response-partition invariant, not the number of experiences, a scalar amount of consciousness, a human quale label, or a full phenomenal metric.

Cross-links	N_1	$N_n, n \geq 2$	Directed pair cycle	Whole-update image size
-------------	-------	-----------------	---------------------	-------------------------

9.2 Proposition: three distinct causal-retention regimes

Cross-links	N_1	$N_n, n \geq 2$	Directed pair cycle	Whole-update image size
(0, 0)	1	1	absent	2
(1, 0)	2	1	absent	4
(0, 1)	2	1	absent	4
(1, 1)	4	4	present	8

Proof. For (0, 0), the pair matrix is zero. With exactly one cross-link it has rank one and square zero. With both cross-links it is the swap matrix and has square equal to the identity; all its powers are invertible. Rank-nullity gives the values of N_n . At fixed input u , the whole transition first determines $s^+ = s \oplus u$, so the images from the two different current s values are disjoint. For each such value the pair map has $2^{\text{rank } A}$ images, giving 2, 4, 4, 8. The pair has a directed two-edge cycle precisely when both cross-links are present. $\square$

Consequently, adding a first link creates one-step distinguishability of some initial pair perturbations; adding the complementary link creates indefinitely sustained distinguishability in this exact noiseless model. Neither single link alone produces that sustained response property. Nonproduct organization, recurrence, and persistence are therefore different properties.

9.3 Conditional experiential interpretation

Image cardinality is preserved under bijective state conjugacy. The settings with image sizes 2, 4, and 8 therefore have different complete modeled transition types. If these are actual tokens and the stipulated organization is complete for them, C1-OI entails different whole-token experiential types. C1 says more specifically that their distinctions, causal relations, and temporal-retention relations belong to the same tokenwise organization in experiential presentation.

Throughout the family the implemented s -process has the same current state, update, and represented interfaces under common s_0, u . C1-W consequently gives its type equivalence under the same actuality/completeness qualifications.

9.4 Shared physical example, not an independent proof of identity

The mathematics uses specified dynamics, boundary conventions, and admissible resets. It does not require U1 or C1 to calculate N_n . UCT supplies a conditional experiential interpretation of independently specified organizational facts; the numbers do not independently verify that interpretation. An output-only whole-token equivalence hypothesis identifies all four settings, while UCT's complete-organization hypothesis separates the three image-size classes under its stated premises. This is a difference of conditional theoretical commitments, not an observed phenomenal discrimination or a refutation of every functionalist account.

UCT II v1.1 uses the same family to separate causal retention, graph recurrence, and unchanged declared consumer output, while retaining target-specific semantic and biological bridges. Nilpotence, invertibility, and response partitions are established mathematical tools; no priority claim is made for that mathematics or for the general possibility of behaviorally silent internal change.

10. Scientific bridge architecture

C1-OI concerns complete ontic organization and complete experiential structure.

Ordinary science observes neither completely.

Finite empirical tests therefore require explicit auxiliary structures.

10.1 Declared phenomenal target

Let:

$$\Psi_v(P)$$

be a finite, preregistered phenomenal/psychophysical target inferred from some combination of:

- similarity judgments;
- discrimination performance;
- subjective ratings;
- confidence;
- no-report perceptual indicators;
- multidimensional report patterns;
- physiological measures with independently justified interpretation.

By default:

$$\boxed{\Psi_v(P) \neq \Phi(P).}$$

It is a scientific projection, not complete phenomenology.

10.2 Sufficiency and reflection are different bridge hypotheses

Let D_v be a frozen finite physical view and Ψ_v a declared finite phenomenal target. A **sufficiency bridge** $B_v^{\rightarrow}$ asserts

$$D_v(A) \cong D_v(B) \implies \Psi_v(A) \cong \Psi_v(B).$$

Equivalently, on the compared classes, the target factors through a map f_v from the physical view. This does not require that f_v be injective. To assert a biconditional one must separately impose a **reflection bridge** $B_v^{\leftarrow}$:

$$\Psi_v(A) \cong \Psi_v(B) \implies D_v(A) \cong D_v(B).$$

Together these bridges say that the physical view and the finite target distinguish the same classes within the declared scope. C1-OI for complete structures does not imply reflection after either side has undergone a lossy projection. For example, even an identity correspondence between two full physical and phenomenal states can have an observed target that assigns both states the same response. The finite target then remains sufficient as a function of the physical state but fails reflection.

10.3 Operational Structural Correspondence

Two packages should therefore be distinguished:

$$\begin{aligned} OSC_v^{\rightarrow} &= C1 + B_v^{\rightarrow} + \text{measurement assumptions,} \\ OSC_v^{\leftrightarrow} &= C1 + B_v^{\rightarrow} + B_v^{\leftarrow} + \text{measurement assumptions.} \end{aligned}$$

The notation denotes packages of assumptions, not numeric addition. The operational implications are supplied by the bridge conditions; attaching C1 explains their theoretical interpretation and does not constitute a separate empirical derivation of those conditions. The one-way package predicts target equivalence for equivalent frozen physical views. The two-way package additionally predicts target non-equivalence for non-equivalent views. Reflection must be defended for the specific target; it cannot be supplied merely by requesting more confidence in the same coarse report.

Approximate applications must declare physical and phenomenal distance functions, equivalence tolerances, and error models. Failure to reject a measured difference is not evidence of equivalence without a suitable equivalence-testing design and adequate precision. The exact finite examples below do not solve this measurement problem.

A finite empirical failure challenges the operational package. It does not automatically isolate C1 from either bridge direction or from measurement assumptions.

11. Physical Token/View Selection Protocol

The main falsifiability danger for UCT is post-hoc redefinition.

If a failed experiment can always be answered by choosing a new token, scale, horizon, intervention family, or structural descriptor after seeing the phenomenal result, the theory becomes scientifically non-vulnerable.

Version 1.1 therefore proposes the Physical Token/View Selection Protocol (PTVSP).

Before phenomenal unblinding:

1. declare candidate actual-token hypotheses;
2. declare admissible physical views;
3. declare physical intervention families;
4. declare held-out physical response targets;
5. evaluate physical predictive sufficiency;
6. evaluate stability under measurement perturbation;
7. evaluate cross-system reproducibility when possible;
8. prefer parsimonious physically sufficient views or a preregistered Pareto set;
9. freeze token/view/bridge definitions;
10. only then join or reveal phenomenal outcomes.

The central rule is:

No Post-Hoc Token/View Rescue.

A failed preregistered bridge remains failed.

Changing token, view, grain, horizon, or bridge afterwards creates a new version of the hypothesis.

12. Empirical failure conditions

12.1 Same frozen physical view, different phenomenal target

If

$$D_v(A) \cong D_v(B), \quad \Psi_v(A) \not\cong \Psi_v(B),$$

then the frozen one-way operational bridge fails, subject to its error model. It would be circular to retain that very implication as an independently secure premise while using its counterexample to isolate C1. The failure directly concerns the view/target bridge, not automatically the full identity axiom.

A countermodel explains the distinction: take two different complete physical structures, identify each tokenwise with its experiential structure, let the finite physical view map both to one value, and let the finite phenomenal target distinguish them. C1 is satisfied, while the proposed finite sufficiency bridge fails. Thus finite physical underdescription alone can produce this pattern.

A challenge specifically to C1-W would instead require independently supported equivalence of the **complete constitutive** physical organizations and a valid phenomenal observation that cannot differ between isomorphic full experiential structures. Those are stronger premises than agreement on a frozen coarse view. They must not be smuggled in by renaming the failed finite prediction “completeness.” In ordinary experiments the operational package is the direct test target.

12.2 Different frozen physical views, identical phenomenal target

Under a declared two-way operational bridge, the pattern

$$D_v(A) \not\cong D_v(B), \quad \Psi_v(A) \cong \Psi_v(B)$$

refutes the finite reflection direction. The same full-identity construction with an information-preserving physical view and a constant phenomenal readout satisfies C1 while failing reflection. A difference can therefore be discarded by measurement even when it belongs to the full structure.

To challenge the converse full-structure implication, one would need independently supported complete physical nonisomorphism **and** complete experiential equivalence, not merely equivalent reports. This paper does not claim that routine measurements supply those premises.

C1 is not needed to derive operational predictions that have already been stipulated in a bridge. A successful bridge test supports that fixed bridge/model against specified alternatives; it is not, solely for that reason, differential evidence for C1 over every other interpretation using the same operational law. Any stronger model comparison must expose independently motivated constraints that generate different likelihoods or predictions. This is a statement about empirical discrimination, not an additional foundational mechanism of experience.

12.3 Behaviorally matched causal rewiring

A particularly informative design would match:

- overt behavior;
- report;
- task accuracy;
- confidence;
- access variables;

while manipulating a preregistered constitutive causal relation that survives the frozen physical view.

For actual tokens whose complete constitutive structures differ, C1-OI predicts a difference in complete experiential organization even when the chosen output variables match. A finite measurement difference additionally requires the corresponding reflection bridge. If every available phenomenal indicator has been matched by design, the remaining structural claim has not thereby become an observed empirical discrimination.

This places UCT within the empirical pressure created by unfolding/substitution arguments against causal-structure theories (Doerig et al., 2019; Kleiner, 2020b; Tsuchiya et al., 2020; Usher, 2021; Herzog et al., 2022; Basri, BarAvi, & Hemmo, 2026).

12.4 Latent-memory test

Match explicit current report and task content while preserving a preregistered difference in latent constitutive memory dispositions that changes future admissible response.

C1-OI predicts a difference in complete experiential organization, though not necessarily a difference in explicitly recalled content.

This is a conditional consequence of the identity commitment; other sufficiently strong causal-structural accounts may share it. A distinguishing experiment needs a named competing prediction and a separately justified sensitive phenomenal target. Internal-model verification alone does not supply either.

13. Relation to UCT II and neighboring theories

UCT II asks whether operational cores of major consciousness theories can be reconstructed, embedded, reinterpreted, or shown to conflict as effective organization theories on top of the UCT mother formalism.

The v1.1 interface is explicitly view-based:

$$D_v^{PA}(P) := \Pi_v[\mathbf{D}^{ontic}(P)], \quad O_T = \mathcal{J}_T(D_v^{PA}(P), B_T).$$

An estimate is denoted $\widehat{D}_v^{PA}$, and an underidentified view yields an output set rather than an arbitrarily selected point. The tuple formerly written $\mathfrak{D}^{PA}$ in UCT II v1.0 is a physically anchored scientific representation when used in finite translation, not a license to identify every chosen view with complete ontic organization.

Reproducing target-theory computations requires fixed bridge conventions, not C1 or U1. No-universal-gate relocation additionally uses U1 and a scoped nonuniversality witness. Interpreting full physical organization experientially uses C1. Finite phenomenal prediction needs a separate operational bridge. The shared example in Section 9 respects these distinctions.

The relation is asymmetric:

$$\boxed{\text{failure of one UCT-II bridge} \not\Rightarrow \text{failure of C1 or the process ontology}}$$

but:

$$\boxed{\text{failure of C1} \Rightarrow \text{revision of the UCT-specific interpretation of UCT II.}}$$

In the v1.2 dependency architecture, Paper II's former references to A1, A3, and A5 migrate to U1, P3, and C1 respectively. BAC/TFR, the F/V axes, and the formal reconstruction machinery remain downstream and do not provide evidence that proves C1.

UCT II introduced explicit Bridge Admissibility Conditions, Translation Fidelity Requirements, formal recovery classes, and separate empirical-validation levels. Those constraints remain essential for preventing a rich mother formalism from trivially "containing" other theories by lookup.

Version 1.2 does not claim novelty for:

- panexperientialism or panpsychism;
- process-oriented metaphysics of experience;
- mind-brain or mind-physical identity strategies in general;
- causal/phenomenal structuralism;
- multiscale process accounts;
- overlapping conscious systems;

- mechanistic physical implementation criteria;
- structural approaches to subjectivity;
- the general idea that theories may be partially complementary.

Several neighboring traditions overlap with individual UCT commitments. Panexperientialist and process traditions already treat experience as widespread or fundamental rather than as a late biological product (e.g., Rosenberg, 2004, building on earlier process traditions). IIT explicitly advances an explanatory identity between phenomenal structure and a substrate’s intrinsic cause–effect structure (Albantakis et al., 2023), while neuropsychophenomenal structuralism develops structural-similarity constraints and relational accounts of specific phenomenal character (Fink, Kob, & Lyre, 2021; Lyre, 2022). General mind–brain identity theories likewise deny that mental and physical processes are merely correlated. UCT therefore does not claim to originate universal experience, identity theory, phenomenal structuralism, or structure-preserving mappings.

The combination, de-combination, overlap, and individuation problems are also established literatures (Chalmers, 2016; Miller, 2018; Blackmon, 2021). Recent preprint-level structural-subjectivity programs go further by proposing positive structural criteria for the presence or locus of subjectivity; for example, Nagae (2026) identifies subjectivity with viability-constrained irreversible exclusive commitment. UCT differs at this point: it does not use action, viability, irreversibility, recurrence, access, self-modeling, or any other nonuniversal organization as a basal experience-existence gate, and it presently offers only a scoped no-free-selection theorem rather than a universal positive subject criterion.

Neighboring and critical literatures also include Information Closure Theory (Chang et al., 2020), universal-theory criteria (Kanai & Fujisawa, 2024), mechanistic computation and implementation accounts (Piccinini, 2015; Anderson & Piccinini, 2024), and the unfolding/substitution debate (Doerig et al., 2019; Kleiner, 2020b; Tsuchiya et al., 2020; Usher, 2021; Herzog et al., 2022; Basri et al., 2026).

Recent structural work narrows the originality claim further. Kleiner (2020a) formalizes experience spaces using equivalence classes induced by automorphisms; Kleiner and Ludwig (2024) analyze what it means for conscious experience itself to instantiate a mathematical structure and explicitly study topological structures; Oizumi, Lim, and Kanai (2026) use principal-bundle geometry, symmetry orbits, and a quotient space to characterize perceptual qualia in systems whose consciousness is independently assumed. Those works show that **equivalence classes, topology, quotient constructions, and structural mappings are not distinctive UCT inventions.**

The v1.2-specific role of $\mathfrak{S}_{\mathcal{K}}$ is correspondingly narrow: it is a bookkeeping space of isomorphism classes used to express the consequence that, under C1, the physical and experiential descriptions of the same token trace the same structural-type path. This does not compete with a principal-bundle theory of perceptual content and does not independently solve the problem of selecting the empirically correct topology.

The contribution claimed here is therefore not any single ingredient. Version 1.2 reorganizes the dependency architecture around one tokenwise identity axiom while retaining established mathematics for products, quotient spaces, symmetry, and causal abstraction. The constrained conjunction is now:

actual-process ontology with persistence and lineage
 +one tokenwise Structural-Experiential Identity axiom (C1)
 +derived universal nonempty experience (U1)
 +derived structural continuity on $\mathfrak{S}_{\mathcal{X}}$ (U2)
 +composition under tokenwise structural identity
 +experience/subject separation
 +restricted selection obstruction in an overlap orbit
 +explicit ontic/view/estimate firewall
 +failure-prone finite bridge methodology.

The earlier targeted prior-art audit did not establish historical priority for this conjunction. The axiom minimization itself is a dependency clarification, not a new fundamental mathematical law. Product-preservation, quotient-topology reasoning, orbit symmetry, causal abstraction, and behaviorally silent internal change all have established mathematical or conceptual antecedents. Scientific value must therefore come from the restrictions imposed by the package and from independently constrained applications.

The closest structural antecedents nevertheless differ in scope. Neurophenomenal structuralism constrains mappings between phenomenal and neural relational structure for conscious contents; Kleiner’s formal work treats mathematical structures and equivalence classes of experience; the principal-bundle account organizes perceptual qualia by neural symmetry; IIT identifies experience with a maximally irreducible Φ -structure. UCT’s distinctive commitment, if retained, is instead **universal tokenwise applicability to every actual valid process**, combined with process-token plurality and the refusal to use maximality, recurrence, access, or selfhood as universal existence gates. This is a difference in ontological scope and dependency architecture, not a claim to have invented structural methods.

The prior-art boundary is narrower still when panpsychist and neutral-monist traditions are included. Panpsychist and panexperientialist discussions have long combined ubiquitous experience with event/process ontologies and structural descriptions of physical relations. Russellian monism likewise treats consciousness as intimately connected with the intrinsic nature underlying physical structure and dynamics, although standard Russellian versions typically retain nonstructural quiddities or categorical properties rather than identifying experience with structure alone (Alter & Pereboom, 2023). Nešić (2022) explicitly develops a **neutral-structuralist** theory of consciousness and selfhood from an information-theoretic structural realist starting point and argues that structural accounts require a distinction between extrinsic and intrinsic structure/dynamics. These are close antecedents to parts of UCT’s motivation. UCT therefore does not claim priority for combining panpsychist/panexperientialist scope with structural or neutral-monist language.

The narrower difference claimed for UCT is that C1 makes the **complete token-relative organization itself** the intrinsic experiential organization, without adding a second quiddistic phenomenal variable, while the actual-process ontology permits overlapping and nested experiential tokens and the bridge methodology keeps finite scientific views distinct from the ontic identity claim. Whether that package is ultimately preferable to Russellian or neutral-structuralist alternatives remains an open comparative question.

The same comparison must include recent criticism. Negro (2025) argues against metaphysical phenomenal structuralism while preserving methodological structuralism; Kleiner (2024) argues that the structural turn can proceed independently of metaphysical commitments and cautions against overloading isomorphism; Aldé (2026) presents additional geometrical objections to similarity structuralism. These publication records and DOIs were rechecked against publisher pages during the v1.2 prepublication audit. These works do not straightforwardly refute C1 because UCT’s structural signature is broader than similarity geometry, but they directly challenge any claim that structural equivalence by itself closes the metaphysical explanatory gap.

14. Strong objections and limitations

14.1 The core identity commitment and process ontology are strong

C1 is not an empirical discovery; it is the theory's consciousness-specific identity commitment. The process ontology also makes substantive choices about actual tokens, boundaries, persistence, and lineage.

U1 and U2 are derived results, not additional starting axioms. Their derivability reduces redundancy but does not make C1 empirically established.

The formal theorems must therefore be evaluated partly by the explanatory costs and empirical obligations of C1 and the process ontology.

14.2 C1 risks tautological explanatory rhetoric

Once structural identity is assumed, results of the form:

$$D \neq D' \Rightarrow \Phi \neq \Phi'$$

are straightforward consequences.

They should not be marketed as deep independent discoveries.

The nontrivial content must come from:

- process individuation;
- physical anchoring;
- exact no-go consequences;
- multiscale relations;
- subject non-selection;
- independently constrained bridge predictions.

A stronger challenge is **definition-vacuity**. If $\Phi(P)$ were introduced merely as “an isomorphic copy of $D(P)$ ” with no independently meaningful experiential interpretation, C1 would become a relabeling convention rather than a substantive identity thesis. UCT therefore intends Φ to refer to experiential organization, independently anchored in human cases by phenomenology/psychophysics and connected to finite measurements only through explicit bridge hypotheses. Those bridges can fail. This does not prove C1, but it prevents success from being guaranteed solely by defining the target to equal the physical model.

14.3 Complete ontic organization is not directly observed

Science works with:

$$D_v$$

and:

$$\widehat{D}_v.$$

Direct empirical isolation of full C1-OI from bridge assumptions is therefore difficult.

UCT should not disguise this epistemic limitation.

14.4 Complete phenomenology is not directly measured

Human report, behavior, psychophysics, and physiology are projections.

Even convergent measures do not automatically equal complete phenomenal structure.

This limits the strength of empirical falsification claims.

14.5 Process proliferation

A permissive process ontology can yield many weak, overlapping, or nested experience-bearing tokens.

UCT accepts such plurality as part of its intended ontology. The number of tokens is not a problem to be eliminated by an arbitrary experience threshold; the scientific task is to distinguish their organizations and relevance to a specified target.

Scientific salience and experience existence are separate questions.

This plurality also means UCT does not solve the classical panpsychist subject-summing problem by claiming that micro-subjects literally add up to one macro-subject. An actual macroprocess receives its own experiential organization directly under C1 if it is itself a valid token; continuing microprocesses may simultaneously remain experiential. This avoids making macroexperience a sum of microexperiences, but at the cost of accepting abundant overlapping experiential processes.

The token criteria reduce arbitrary redescription, but they do not prove a unique privileged carving of every physical history. A scientific application must therefore justify why its selected token is the relevant physical process for the target question rather than treating any convenient connected subhistory as automatically privileged.

14.6 Macroprocess descriptions are applications, not new existence gates

UCT accepts actual causal processes at different scales. The finite criterion in Section 2.5 makes that domain concrete in one classical representation. No unique natural macro boundary, autonomous coarse state, or maximal integration score is required before an actual process can possess experience. In concrete applications, competing claims about physical boundaries and constitutive relations remain accountable to the system being described. Uncertainty in that physical identification is not evidence that the framework lacks a separate macro-experience mechanism.

14.7 Subject terminology must not become a prerequisite

If subject names an experience-bearing process, the ontology supplies many possible referents. If it names a conceptual owner, self-model, or one exclusive human-like agent, additional organizational descriptions are relevant to that usage. UCT need not add an extra owner of experience to explain why processes have experience. The optional selection theorem constrains exclusive labeling rules; failure to specify one universal subject partition is not a foundational incompleteness criterion adopted by this paper.

14.8 General quality identity and biological specification

C1 supplies the same general account of experiential quality for elementary processes, cells, animals, and humans: the quality is the intrinsic presentation of the actual organization. Human complexity does not introduce a second explanatory gap inside the theory. The process ontology plus C1/U1/U2 do not provide a biological lookup table assigning familiar labels such as red or unpleasant to every measured neural configuration. Obtaining such mappings is a task for particular scientific applications, just as identifying a concrete cellular organization is. It must not be misreported as absence of the general physical-experiential principle.

14.9 Cross-species qualia identity is underidentified

Similar behavior, homologous mechanisms, or structurally comparable organization do not by themselves prove identical subjective qualities across species.

Claims should therefore use terms such as:

- precursor;
- conserved mechanism;
- structural homology;
- convergent similarity;

rather than asserting uncalibrated qualitative identity.

14.10 Identity, isomorphism, and explanatory strength

C1 uses an isomorphism as the formal expression of a stronger metaphysical identity claim. The mathematics alone establishes only structural equivalence; it does not prove that the physical and experiential descriptions are numerically or metaphysically identical. This distinction is important because IIT also advances an explanatory identity between experience and a physically characterized structure, while explicitly distinguishing identity from mere isomorphism.

UCT's position is therefore conditional and substantive: the same token-relative organization is asserted to admit physical and intrinsic experiential presentation, and h_P represents that commitment in the declared structural language. A critic can reject this identity interpretation while accepting all of the structural mathematics. The theory must not cite the existence of an isomorphism as an independent proof of the metaphysical identity that licensed it.

14.11 Metaphysical structuralism objections remain substantive

C1 is stronger than methodological structuralism: it is a metaphysical identity commitment. Recent critiques of metaphysical phenomenal structuralism therefore matter directly to UCT rather than being dismissed as attacks on a different research program.

Negro (2025) argues, among other things, that relational structure may fail to determine intrinsic phenomenal character and presses a “lonely experience” objection against views requiring an experience to be fully constituted by its relations to other experiences. Aldé (2026) develops further structural objections using color geometry. Kleiner (2024) separately emphasizes that structural methods in consciousness science can be valuable without settling the underlying metaphysics and cautions against treating isomorphism itself as the explanation of phenomenal structure.

UCT is not identical to **similarity structuralism**. Its declared structure can contain pointed state, intrinsic state distinctions, causal/temporal operations, intervention ports, constituent relations, and even a one-element experiential carrier; it does not require every experience to be individuated only by pairwise similarity relations to other experiences. Thus the “one experience requires at least two relata” objection is not automatically decisive against C1.

Nevertheless, the broader intrinsic-character challenge remains. C1 asserts that the complete token-relative organization is also the intrinsic experiential organization; it does not derive that identity from structural mathematics alone. The objections therefore remain genuine pressure on C1's metaphysical plausibility and should be treated as such, not as already solved by the formalism.

14.12 Russellian and neutral-structuralist alternatives

Russellian monism supplies a particularly important alternative to C1. It can accept that physics primarily captures structure and dynamics while holding that nonstructural categorical or quiddistic properties underlie that structure and are relevant to consciousness.

On such a view, complete structural knowledge need not exhaust intrinsic phenomenal nature.

UCT deliberately rejects that extra hidden phenomenal degree of freedom under fixed complete token-relative organization. This is a substantive disagreement, not a terminological difference. The No Hidden Quale result follows only inside UCT's identity framework; it is not a neutral theorem that rules out Russellian quiddities.

Neutral-structuralist proposals such as Nešić (2022) create a different pressure: one may preserve structural explanation while avoiding either straightforward panpsychism or strict physicalist structural identity. UCT therefore bears the burden of showing that its universal tokenwise identity gives explanatory or empirical advantages over these alternatives rather than merely restating a structural vocabulary.

14.13 Topology underdetermination

C1 fixes equality of physical and experiential **isomorphism classes** for the same token, but it does not uniquely determine a topology, metric, or notion of approximate structural similarity between different classes. U2 therefore has the logical form:

predeclared comparison geometry+physical path continuity+ $C1 \Rightarrow$ experiential structural continuity.

It does not have the stronger form:

$C1 \Rightarrow$ nature supplies one unique consciousness metric or continuous evolutionary path.

This underdetermination is a scientific obligation rather than a contradiction. Different application domains may require different physically justified invariant metrics, and their adequacy must be assessed prospectively rather than chosen to preserve UCT after a failed comparison.

14.14 Relativistic and quantum extensions

The present finite/classical formalism is not a final fundamental-physics theory.

A relativistic version should be formulated over causal spacetime histories rather than a preferred global time slice.

A quantum version requires appropriate state spaces, channels/instruments, subsystem criteria, and intervention notions.

14.15 Semantic representation

Causal tracking does not automatically establish semantic aboutness.

Detailed models of semantic representation, conceptual content, and explicit self-reference belong to higher-order applications. They do not supply prerequisites for basal experience or a second physical-experiential principle.

14.16 Axiom minimization is not independent evidence

A smaller independent axiom basis can improve clarity, dependency control, and explanatory economy, but it is not by itself evidence that C1 is true. The move from several separately stated consciousness commitments to one stronger tokenwise identity commitment reduces redundancy while concentrating rather than eliminating the metaphysical burden.

Accordingly:

logical compression $\neq$ empirical confirmation.
--

Any empirical advantage must come from prospectively constrained bridge predictions, comparative explanatory success, or failed alternatives—not from counting the number of displayed axioms.

14.17 Formal strength and finite measurements must remain explicit

C1 is a strong interpretation of identity. Its tokenwise isomorphisms support the product proof; the weaker C1-OI formula by itself does not. The present candidate preserves this difference rather than attributing the stronger formal statement retroactively to a weaker pairwise formula. A critic may reject the identity commitment; that is distinct from an internal derivation failing to follow from it.

Finite scientific measurements can discard distinctions even when complete experiential and physical organizations share the same structure. Therefore finite reflection remains an additional condition of an operational two-way test, not another experience-generating mechanism or another foundational axiom. The measurement qualifications do not replace the unified explanation of experience.

15. Claim-status summary

Claim	Status in v1.2
Actual valid process-token domain	Process ontology
Persistence under embedding	P3 process-ontology rule
Token/type/lineage distinction	P6 process-ontology definition
Tokenwise structure-for-structure identity	C1 — sole consciousness-specific core axiom
Universal experience for actual valid process tokens	Theorem U1 , derived from nonempty actual organization + C1
Scoped continuity of experiential organization	Theorem U2 , derived on the common structural-type space
Unique topology/metric on the structural-type space	not supplied by C1; application-level structural geometry must be justified
Human-like conceptual self not required for basal experience	Corollary U3 , given a non-self actual-token witness
Pairwise type equivalence	C1-OI consequence
One-way phenomenal completeness	C1-W consequence
Organization-Gate Impossibility	theorem conditional on U1 + nonuniversality witness
No Hidden Quale	consequence of C1-W
Probe Invariance	consequence of ontic/probe separation + C1-W
No Future Contamination	consequence of C1-W
Causally Screened History	consequence of C1-W + P6
Nested coexistence	P3 + U1
Non-Summative Combination	actual tokens + C1 + standard product in the declared common category
Extra experience-generating composition mechanism	not introduced
C1-OI alone transports independent products	false; full tokenwise C1/product typing is needed
Macro invariance with micro change	macro equivalence: C1-W; lower-type difference: C1-OI

Claim	Status in v1.2
Symmetric-overlap non-selection	orbit lemma plus explicit nonempty/exclusive candidate restriction
Intrinsic structure rules out every invariant partition	not established
Canonical positive subject selector	not a required universal element of UCT
General account of human/cellular quality	one C1 identity principle
Familiar quality labels mapped to measured mechanisms	application-level biological/psychophysical task
Structural evolution from physical process to conceptual self	explanatory/application synthesis using U1/U2 + physical path premise
No First Conscious Ancestor	Corollary E1 from U1: every actual valid token has nonempty experience
Evolutionary continuity proves C1	explicitly rejected
First conscious ancestor inside UCT	none; Corollary E1 follows from U1 for every actual valid token
Continuity alone proves universal nonempty experience	false; smooth zero-to-positive countermodels exist without C1/U1
If experience existence itself is continuous on a connected ancestor spectrum	Conditional Lemma E3: the binary existence predicate is constant; a conscious human endpoint fixes it to 1 throughout
Perfect mathematical continuity required for the evolutionary intuition	false; Corollary E2 transfers arbitrarily fine physical structural chains to equally fine experiential chains
“Ancestor” from particles to humans is literal Darwinian ancestry	false; prebiotic/elementary stages are a schematic physical/chemical prehistory joined to biological evolution
Fewer displayed axioms independently confirm C1	false; minimization is dependency cleanup, not empirical evidence
PTVSP	methodological proposal
OSC _v	finite operational bridge package, one-way/two-way forms distinguished
Finite bridge reflection	additional requirement for a two-way empirical package
Three-register/four-setting examples	exact finite mathematical thought experiments; no phenomenal validation
C1 empirically established	not established
A smaller axiom list empirically confirms UCT	explicitly rejected
Hard problem deductively solved for all critics	not claimed
Extra universal subject owner or partition	not postulated as a prerequisite
Panexperiential + structural + process-ontology combination is wholly novel	false; close antecedents exist in panpsychist/process, Russellian, and neutral-structuralist traditions
Historical priority of the full package	not established

16. Conclusion

UCT I can be stated more economically than in v1.1 without changing its central metaphysical picture. The process ontology identifies actual tokens, persistence, boundaries, types, and lineages. One consciousness-specific commitment, C1, identifies each actual token’s complete physical organization with its intrinsic experiential organization in a declared common structural signature.

From C1 and the process ontology, Universal Nonempty Experience becomes Theorem U1 rather than an independent axiom. When physical and experiential descriptions are placed

in the same isomorphism-class space with a predeclared invariant topology or pseudometric, Structural Continuity becomes Theorem U2 because the two quotient-valued trajectories are the same function. C1 does not uniquely choose that topology or prove that a real evolutionary trajectory is continuous within it; those are separate structural-modeling obligations. Human-like conceptual selfhood is then a higher-order organization rather than a basal prerequisite. Persistence under embedding and token/type/lineage remain substantive, but as process-ontological rules rather than additional consciousness axioms.

This reclassification reduces logical redundancy; it does not make the theory empirically established. C1 remains a strong metaphysical identity commitment. Science accesses finite physical views and phenomenal/behavioral targets, not complete ontic organization directly. Prospective bridge assumptions, token/view freeze rules, and explicit failure conditions remain necessary.

Evolutionary continuity acquires a cleaner and more positive role. Within UCT there is no first conscious ancestor. U1 assigns nonempty experience to every actual valid process token, including the simplest actual physical interaction processes. U2 transfers strict structural continuity when such a path is independently continuous, while Corollary E2 transfers arbitrarily fine physical structural chains to equally fine experiential chains without requiring mathematically perfect continuity. Evolution may introduce thresholds for memory, recurrence, global access, self-modeling, language, metacognition, and conceptual selfhood, but these are transitions within experiential organization rather than an ignition of experience itself. The all-ancestors thought experiment makes this no-ignition picture explicit and places an explanatory burden on competing onset theories; it is not used as an independent proof of C1.

The resulting architecture is therefore:

Process Ontology + one Structural-Experiential Identity axiom + derived theorem hierarchy + explicit

The next scientific burden is not to add more existence axioms, but to identify actual organization faithfully, construct finite bridges that can fail, and compare the resulting predictions with neighboring theories under fixed protocols. Version 1.2 freezes the foundational architecture and moves the project to author review and release decision rather than further axiom expansion.

16.1 Cross-version compatibility with UCT II v1.1

UCT II v1.1 remains interpretable without rerunning its formal toy calculations. Its UCT-I dependencies migrate as follows:

UCT II v1.1 dependency	UCT I v1.2 interpretation
A1 Universal Experience	U1 Universal Nonempty Experience
A3 Persistence Under Embedding	P3 Persistence, cessation, and embedding
A5-SI / A5	C1 Structural-Experiential Identity
A5-OI	C1-OI
A5-W	C1-W

BAC, TFR, the F/V classification, and the view-based reconstruction machinery remain downstream methodological structures. Their formal reconstruction results do not become evidence for C1 merely because the foundational labels have changed.

No claim is made here that UCT II v1.1 has itself been version-updated. A future paired revision may update labels and dependency notes while preserving its released v1.1 record.

References

Reference and version note. Existing v1.1 references are retained for citation continuity. No identifier has been assigned to version 1.2, and publication remains paused. Source use does not imply empirical support for C1.

Reference-status note. Literature added specifically for the v1.1 evolutionary/comparative synthesis was checked against publisher, PubMed, or journal records during the 29 September 2026 claim audit. Basri et al. (2026) was indexed by PubMed as ahead-of-print on 8 September 2026 even though the journal issue is dated November 2026. The prior-art audit also added Rosenberg (2004) as an explicit panexperientialist comparison and Nagae (2026) as a recent preprint-level structural-subjectivity comparator. The v1.2 prior-art update additionally verified Kleiner (2020a), Kleiner (2024), Kleiner and Ludwig (2024), Fink, Kob and Lyre (2021), Lyre (2022), Nešić (2022), Alter and Pereboom (2023), Negro (2025), Aldé (2026), Anderson and Piccinini (2024), and Oizumi, Lim and Kanai (2026) against journal/publisher or authoritative bibliographic records. Oizumi et al. has a 29 September 2026 production correction affecting supplementary material and Figure 4 orientation; the main article DOI remains pgag261. The correction does not alter the relational/symmetry/quotient-space point for which the article is cited here.

Aldé, D. (2026). Two geometrical arguments against similarity structuralism. *Synthese*, 207, 204. doi:10.1007/s11229-026-05562-5.

Alter, T., & Pereboom, D. (2023). Russellian Monism and Structuralism About Physics. *Erkenntnis*, 88, 1409–1428. doi:10.1007/s10670-021-00408-7.

Albantakis, L., Barbosa, L., Findlay, G., Grasso, M., Haun, A. M., Marshall, W., Mayner, W. G. P., Zaeemzadeh, A., Boly, M., Juel, B. E., Sasai, S., Fujii, K., David, I., Hendren, J., Lang, J. P., & Tononi, G. (2023). Integrated information theory (IIT) 4.0: Formulating the properties of phenomenal existence in physical terms. *PLoS Computational Biology*, 19(10), e1011465. doi:10.1371/journal.pcbi.1011465.

Anderson, N. G., & Piccinini, G. (2024). *The Physical Signature of Computation: A Robust Mapping Account*. Oxford University Press. doi:10.1093/9780191872075.001.0001.

Auvray, M., Myin, E., & Spence, C. (2010). The sensory-discriminative and affective-motivational aspects of pain. *Neuroscience & Biobehavioral Reviews*, 34(2), 214–223. doi:10.1016/j.neubiorev.2008.07.008.

Basri, J., BarAvi, P., & Hemmo, M. (2026). Functionalist theories of consciousness: unfolding, substitution and the indeterminacy of function by the physics of the brain. *Consciousness and Cognition*, 145, 104121. Advance online publication, 8 September 2026. doi:10.1016/j.concog.2026.104121.

Beckers, S., & Halpern, J. Y. (2019). Abstracting causal models. *Proceedings of the AAAI Conference on Artificial Intelligence*, 33(01), 2678–2685. doi:10.1609/aaai.v33i01.33012678.

Bielecki, J., Dam Nielsen, S. K., Nachman, G., & Garm, A. (2023). Associative learning in the box jellyfish *Tripedalia cystophora*. *Current Biology*, 33(19), 4150–4159.e5. doi:10.1016/j.cub.2023.08.056.

Blackmon, J. C. (2021). Integrated Information Theory, intrinsicity, and overlapping conscious systems. *Journal of Consciousness Studies*, 28(11–12), 31–53. doi:10.53765/20512201.28.11.031.

Boisseau, R. P., Vogel, D., & Dussutour, A. (2016). Habituation in non-neural organisms: evidence from slime moulds. *Proceedings of the Royal Society B*, 283(1829), 20160446. doi:10.1098/rspb.2016.0446.

Buchholz, P. (1994). Exact and ordinary lumpability in finite Markov chains. *Journal of Applied Probability*, 31(1), 59–75. doi:10.2307/3215235.

Chalmers, D. J. (2016). The combination problem for panpsychism. In G. Bruntrup & L. Jaskolla (Eds.), *Panpsychism: Contemporary Perspectives* (pp. 179–214). Oxford University Press. doi:10.1093/acprof:oso/9780199359943.003.0008.

- Chang, A. Y. C., Biehl, M., Yu, Y., & Kanai, R. (2020). Information Closure Theory of Consciousness. *Frontiers in Psychology*, 11, 1504. doi:10.3389/fpsyg.2020.01504.
- Coninx, S. (2024). Pain philosophy: Recent debates and future challenges. *Philosophy Compass*, 19(4), e12981. doi:10.1111/phc3.12981.
- Doerig, A., Schurger, A., Hess, K., & Herzog, M. H. (2019). The unfolding argument: Why IIT and other causal structure theories cannot explain consciousness. *Consciousness and Cognition*, 72, 49–59. doi:10.1016/j.concog.2019.04.002.
- Doerig, A., Schurger, A., & Herzog, M. H. (2021). Hard criteria for empirical theories of consciousness. *Cognitive Neuroscience*, 12(2), 41–62. doi:10.1080/17588928.2020.1772214.
- Fink, S. B., Kob, L., & Lyre, H. (2021). A structural constraint on neural correlates of consciousness. *Philosophy and the Mind Sciences*, 2, 7. doi:10.33735/phimisci.2021.79.
- Giurfa, M., Zhang, S., Jenett, A., Menzel, R., & Srinivasan, M. V. (2001). The concepts of ‘sameness’ and ‘difference’ in an insect. *Nature*, 410, 930–933. doi:10.1038/35073582.
- Herzog, M. H., Schurger, A., & Doerig, A. (2022). First-person experience cannot rescue causal structure theories from the unfolding argument. *Consciousness and Cognition*, 98, 103261. doi:10.1016/j.concog.2021.103261.
- Jozet-Alves, C., Bertin, M., & Clayton, N. S. (2013). Evidence of episodic-like memory in cuttlefish. *Current Biology*, 23(23), R1033–R1035. doi:10.1016/j.cub.2013.10.021.
- Kanai, R., & Fujisawa, I. (2024). Toward a universal theory of consciousness. *Neuroscience of Consciousness*, 2024(1), niae022. doi:10.1093/nc/niae022.
- Kleiner, J. (2020a). Mathematical models of consciousness. *Entropy*, 22(6), 609. doi:10.3390/e22060609.
- Kleiner, J. (2020b). Brain states matter: A reply to the unfolding argument. *Consciousness and Cognition*, 85, 102981. doi:10.1016/j.concog.2020.102981.
- Kleiner, J. (2024). Towards a structural turn in consciousness science. *Consciousness and Cognition*, 119, 103653. doi:10.1016/j.concog.2024.103653.
- Kleiner, J., & Ludwig, T. (2024). What is a mathematical structure of conscious experience? *Synthese*, 203, 89. doi:10.1007/s11229-024-04503-4.
- Lyre, H. (2022). Neurophenomenal structuralism. A philosophical agenda for a structuralist neuroscience of consciousness. *Neuroscience of Consciousness*, 2022(1), niac012. doi:10.1093/nc/niac012.
- Miller, G. (2018). Can subjects be proper parts of subjects? The de-combination problem. *Ratio*, 31(2), 137–154. doi:10.1111/rati.12166.
- Nagae, M. (2026). *The Structural Conditions of Consciousness VI — Final Volume — Structural Knot Theory: Toward a Structural Decision Tool: Determining the Presence and Degree of Subjectivity*. Zenodo preprint. doi:10.5281/zenodo.20887803.
- Nešić, J. (2022). Towards a Neutral-Structuralist Theory of Consciousness and Selfhood. *International Studies in the Philosophy of Science*, 35(3–4), 243–259. doi:10.1080/02698595.2022.2092825.
- Negro, N. (2025). Three arguments against metaphysical structuralism in consciousness research. *Synthese*, 206, 8. doi:10.1007/s11229-025-05103-6.
- Oizumi, M., Lim, C., & Kanai, R. (2026). Principal bundle geometry of qualia: Understanding the quality of consciousness from symmetry. *PNAS Nexus*, 5(9), pgag261. doi:10.1093/pnasnexus/pgag261. A publisher correction concerning supplementary material and Figure 4 orientation appeared on 29 September 2026 (pgag313).
- Piccinini, G. (2015). *Physical Computation: A Mechanistic Account*. Oxford University Press. doi:10.1093/acprof:oso/9780199658855.001.0001.

- Rosenberg, G. H. (2004). *A Place for Consciousness: Probing the Deep Structure of the Natural World*. Oxford University Press. doi:10.1093/acprof:oso/9780195168143.001.0001.
- Rubenstein, P. K., Weichwald, S., Bongers, S., Mooij, J. M., Janzing, D., Grosse-Wentrup, M., & Schölkopf, B. (2017). Causal consistency of structural equation models. *Proceedings of the 33rd Conference on Uncertainty in Artificial Intelligence*. arXiv:1707.00819.
- Schnell, A. K., Boeckle, M., Rivera, M., Clayton, N. S., & Hanlon, R. T. (2021). Cuttlefish exert self-control in a delay of gratification task. *Proceedings of the Royal Society B*, 288(1946), 20203161. doi:10.1098/rspb.2020.3161.
- Seth, A. K., & Bayne, T. (2022). Theories of consciousness. *Nature Reviews Neuroscience*, 23, 439–452. doi:10.1038/s41583-022-00587-4.
- Tsuchiya, N., Andriellon, T., & Haun, A. (2020). A reply to “the unfolding argument”: Beyond functionalism/behaviorism and towards a science of causal structure theories of consciousness. *Consciousness and Cognition*, 79, 102877. doi:10.1016/j.concog.2020.102877.
- Usher, M. (2021). Refuting the unfolding-argument on the irrelevance of causal structure to consciousness. *Consciousness and Cognition*, 95, 103212. doi:10.1016/j.concog.2021.103212.
- Walters, E. T. (2025). From nociception in aneural animals to human suffering: Toward a comparative biology of pain. *Journal of Experimental Biology*, 228(19), jeb251210. doi:10.1242/jeb.251210.
- Winters, J. J. (2020). The temporally-integrated causality landscape: A theoretical framework for consciousness and meaning. *Consciousness and Cognition*, 83, 102976. doi:10.1016/j.concog.2020.102976.
- Winters, J. J. (2021). The Temporally-Integrated Causality Landscape: Reconciling neuroscientific theories with the phenomenology of consciousness. *Frontiers in Human Neuroscience*, 15, 768459. doi:10.3389/fnhum.2021.768459.
- Yaron, I., Melloni, L., Pitts, M., et al. (2022). The ConTraSt database for analysing and comparing empirical studies of consciousness theories. *Nature Human Behaviour*, 6, 593–604. doi:10.1038/s41562-021-01284-5.

Appendix A. Main changes relative to v1.1

1. Reclassifies the v1.1 axiom list by logical role rather than preserving six parallel foundational labels.
2. Retains one consciousness-specific core commitment:

$$\forall P \in \mathcal{P}, \exists \Phi_{\mathcal{K}}(P), h_P : \mathbf{D}_{\mathcal{K}}^{ontic}(P) \cong \Phi_{\mathcal{K}}(P).$$

3. Derives Universal Nonempty Experience (U1) from actual-token nonempty physical organization plus C1.
4. Introduces the common structural-type space:

$$\mathfrak{S}_{\mathcal{K}} = \text{Obj}(\mathcal{C}_{\mathcal{K}}) / \cong$$

and derives Structural Continuity (U2) by showing that physical and experiential quotient-valued trajectories are the same function.

5. Moves Persistence Under Embedding to process ontology as P3.
6. Moves Token / Type / Lineage to process ontology as P6.
7. Demotes Selfhood Non-Prerequisite to Corollary U3, conditional on an actual non-self token witness.

8. Renames the weaker identity consequences C1-W and C1-OI and preserves their scoped theorem dependencies.
9. Preserves the v1.1 firewall:

$$\mathbf{D}^{ontic} \neq D_v \neq \widehat{D}_v \neq Subject.$$

10. Preserves the Non-Summative Combination theorem, symmetry-orbit result, finite bridge architecture, and Physical Token/View Selection Protocol under updated dependencies.
11. Preserves and incorporates the four-setting causal-retention family, while explicitly separating its mathematics from the experiential interpretation supplied by C1.
12. Recasts evolutionary continuity as an application of U1/U2 plus a physical path premise, not as evidence that independently proves C1.
13. Adds a migration table so references to v1.1 A1-A6 remain traceable.
14. Preserves v1.0 and v1.1 as immutable historical editions and publishes v1.2 only as a linked successor version.
15. Adds an explicit smooth-threshold countermodel showing why evolutionary/structural continuity alone cannot derive universal nonempty experience; C1 remains the required physical-experiential commitment.
16. Re-centers the evolutionary synthesis on Corollary E1 (No First Conscious Ancestor): within UCT, experience is nonempty from the simplest actual processes onward, while evolution transforms experiential organization rather than experience existence.
17. Adds Proposition U0 to state the exact foundational compression, clarifies that minimization is not evidence, and corrects/updates the structural-prior-art bibliography for publication review.
18. Restores the exact three-register circuit figure omitted during migration and promotes the continuity-alone countermodel to a numbered subsection.
19. Performs publication-layout cleanup, including a stable Figure 1 reference for Markdown/PDF portability and a dedicated title-page build.
20. Repairs the All-Ancestors-Survive formalization so U1 applies only to actualized tokens in the stipulated scenario; adds the fine-grained-chain Corollary E2 and the conditional connected-existence Lemma E3.
21. Adds an explicit hostile-review discussion of metaphysical structuralism, definition-vacuity, process proliferation, and the fact that UCT does not solve subject summation by composing micro-subjects.
22. Adds a non-novelty boundary for No First Conscious Ancestor, clarifies elementary-process language, and freezes E2/E3 as the precise graduality/connected-existence results for release review.
23. Version 1.2 freezes the foundational architecture, shortens the abstract, records publisher rechecks for the latest structuralist objections, and prepares the manuscript for final author review without initiating publication.

Appendix B. An exact three-register thought experiment

B.1 Physical specification and scope

Stipulate three physically implemented binary registers s, a, b , a common clock, an external input u , XOR gates, buffered one-way connections from s , and switchable cross-connections between a and b . The switch setting is $\lambda \in \{0, 1\}$. One clock step is:

$$\begin{aligned}
s_{t+1} &= s_t \oplus u_t, \\
a_{t+1} &= s_t \oplus (\lambda b_t), \\
b_{t+1} &= s_t \oplus (\lambda a_t), \\
y_t &= s_t.
\end{aligned}$$

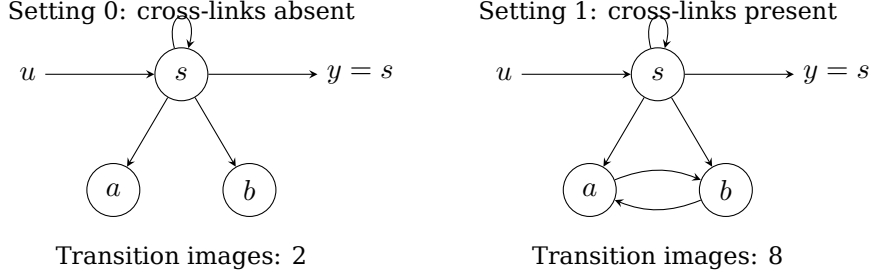


Figure 1: The two stipulated circuit settings. Arrows denote state dependencies. The added pair cross-links distinguish the settings; the external output is the same implemented state in both. Clock and buffering are specified in the text.

Figure 1 displays the two stipulated circuit settings. Multiplication and XOR act on bits. Buffering is stipulated to exclude feedback and loading from the downstream pair into the s update. In real hardware this isolation and all relevant boundary effects would require examination. The mathematics treats the finite update/intervention specification as complete within the stated idealized model. A real circuit's complete ontic structure is not established by this calculation. No circuit was built and no phenomenal observations were collected.

Before calculating any output, specify two token hypotheses: the whole implemented network W_λ , and the continuing implemented register process S_λ with state s , its update mechanism, clock, and declared boundary u . The whole is causally connected through the outgoing s links in both settings. The cross-coupled a, b pair has boundary s . These are explicit component/boundary choices, not subject assignments. U1 applies conditionally if the stipulated processes are actually instantiated and meet the token requirements; a mathematical state table is not itself proof of that instantiation.

The intervention family comprises all 27 coordinate clamps immediately before an update: each register is either unchanged, set to 0, or set to 1. The boundary input takes both bit values. The switch settings are separately compared mechanisms; they are not assumed to constitute a continuous U2 path. This is a worked theoretical specification, not a pre-registered empirical study.

B.2 Exact closure of an implemented sub-process

For $\pi(s, a, b) = s$, the macro update is $\bar{T}_u(s) = s \oplus u$. Every two physical states with the same s have the same next projected state. Projected coordinate clamps act as the same clamp on s if it is targeted and as the identity otherwise. Hence:

$$\pi(T_u^\lambda(Ix)) = \bar{T}_u(\omega(I)\pi(x))$$

for every declared clamp, input, setting, and state. The implementation of s is part of the physical specification, not inferred from closure alone. The present state and future update of this sub-process do not depend on a or b , even though it causally drives them.

By induction on time, for the same s_0 and any input string:

$$y_t = s_0 \oplus u_0 \oplus \cdots \oplus u_{t-1} \quad (t \geq 1)$$

in both settings, independently of initial a, b . This is equality of the declared output behavior, not equivalence under every conceivable additional physical experiment.

B.3 Full structural difference is invariant to state relabeling

Fix either input value. At $\lambda = 0$, both downstream next states equal s , so the eight-state transition map has exactly two images. At $\lambda = 1$, the map is bijective: from (s', a', b') , recover:

$$s = s' \oplus u, \quad a = b' \oplus s, \quad b = a' \oplus s.$$

It therefore has eight images. Conjugating a transition map by a bijective state relabeling preserves its image cardinality. Consequently the two whole-network transition structures are not isomorphic, even before requiring isomorphisms to preserve the intervention ports.

At a fixed boundary s , the $a \mid b$ split has independent component updates at $\lambda = 0$. At $\lambda = 1$, changing b changes a^+ and changing a changes b^+ ; no product of component-local update maps reproduces this behavior relative to these physical ports.

The two distinctions yield compatible conditional UCT statements:

Fixed comparison	Mathematical result	Conditional UCT interpretation
Whole tokens W_0, W_1	Nonisomorphic full modeled dynamics	C1-OI gives different complete experiential types if the stipulated organization is complete and the tokens actual
Implemented processes S_0, S_1	Same state/update and represented interfaces	C1-W gives equivalent experiential types under the same completeness and actuality qualifications
Output y	Same trajectory for common s_0, u	Output equality does not settle whole-token experiential equivalence
Downstream split $a \mid b$, boundary s fixed	Cross-dependence appears at setting 1	A compositional interpretation uses C1 and the specified constituent product; it does not determine one subject

The coexistence of these conclusions is the point of token-relative completeness. Replacing a whole-process claim by the sub-process claim after a failed test would still violate the freeze protocol. Here both comparisons are specified in advance and reported together.

B.4 Two checks against arbitrary coarse-graining

The feature $q = a \oplus b$ obeys $q^+ = \lambda q$. Its exact algebraic closure does not establish a separate actual q -token. No separate parity register is stipulated. A distributed realization could still be proposed and independently examined; absence of a literal register is not a proof of absence of such a process.

The feature $r = s \oplus a \oplus b$ fails even the deterministic closure test at $\lambda = 0$. With $u = 0$, states $(0, 0, 0)$ and $(1, 1, 0)$ both have $r = 0$, but their next r values are respectively 0 and 1. Thus no deterministic one-state update for r alone exists in that scope. This is a model failure, not a no-experience conclusion.

B.5 What the example does and does not distinguish

An output-sufficiency hypothesis assigning whole-token experiential equivalence whenever every declared u -to- y response agrees conflicts with C1-OI on this model, conditional on the stated actuality/completeness premises. This comparator is deliberately narrow. It is not all functionalism: theories that include internal counterfactual organization need not identify the two whole networks. No IIT 4.0 quantity or GNWT neural model is calculated here; this construction is not reported as an empirical victory over either theory.

The example is a finite specialization and audit of the project’s earlier behaviorally silent internal-change argument. Its value is to put output equality, full nonisomorphism, fixed token boundaries, intervention closure, and the limits of coarse descriptions in one independently checkable specification. It does not claim that the general phenomenon or the underlying mathematics is new.

B.6 Exact verification record

The preserved v1.1 standard-library Python script `verification/check_rc4.py` checks 13 scoped properties. It enumerates all 32 state/input/setting transitions, 128 same-projection closure comparisons, and 864 state-reset intervention cases. It also verifies the finite-set compositional counterexample, the restricted symmetry orbit and invariant global counterpartitions, the lossy-target counterexample, the transition-image invariant, and the coarse-projection examples. Full transition data and results are supplied in the preserved v1.1 package.

A separate preserved script `verification/check_transformations.py` checks the four-setting extension, port covariance, and bridge countermodels. General statements rely on the analytic proofs above; finite enumeration is not a proof about arbitrary physical systems and is not empirical validation of C1.

Appendix C. Revision-specific dependency ledger

The table below reflects the **v1.2 formal dependency audit**. It separates ontological typing, consciousness-specific identity, mathematical lemmas, empirical witnesses, and finite bridge assumptions.

Result / node	Required premises	Scope boundary
C1 Structural-Experiential Identity	actual valid token domain; token-relative $\mathbf{D}^{ontic}$; declared common signature $\mathcal{K}$	sole consciousness-specific axiom; not derived from evolution, report, toy models, or bridge success
U1 Universal Nonempty Experience	C1; actual token with nonempty constitutive carrier	internal consequence; universal-experience content is absorbed into C1 rather than independently proved
U2 Structural Continuity	C1; common structural quotient $\mathfrak{S}_{\mathcal{K}}$; predeclared invariant topology/pseudometric	transfer theorem; does not prove that nature supplies a continuous path or a unique consciousness metric
U3 Selfhood Non-Prerequisite	U1; actual-token witness lacking the declared conceptual-self property	witness-dependent; does not derive a positive self-model mechanism
U0 Foundational Compression	C1; P3; P6; U1; U2; U3 and their typed premises	meta-result about dependency economy, not empirical confirmation

Result / node	Required premises	Scope boundary
Organization-Gate Impossibility	U1; explicit actual nonuniversality witness for the candidate mechanism tokenwise C1	rejects only a universal experience-existence gate
C1-OI pairwise type equivalence		weaker than C1 written alone
C1-W one-way phenomenal completeness	tokenwise C1	only licenses conclusions when complete token-relative organization is fixed
No Hidden Quale	C1-W; complete ontic-structure equivalence	finite/coarse view equality is insufficient
Probe Invariance	C1-W; complete ontic equivalence; probe-choice-only change	a probe that physically changes the token can change experience
No Future Contamination	C1-W; same present complete organization; future-only divergence	present causal dispositions remain part of present organization
Causally Screened History: experiential equivalence	C1-W; same current complete organization; differing history with all current constitutive consequences screened	P6 is not needed for this experiential-equivalence conclusion
Causally Screened History: numerical token distinction	preceding equivalence; P6; lineage/history distinction	P6 is used only for the token/type/lineage addendum
External reward-label non-identity	C1-W; intrinsic organization fixed; external label/convention only	does not identify the physical basis of valence
Nested coexistence	P3 persistence; U1	does not imply canonical subject count or subject merger
Non-Summative Combination	actual whole/constituents; C1; declared standard product; physical nonproduct premise	relative nonproduct organization, not micro-subject summation
Macro experiential invariance	C1-W; complete macro ontic equivalence	does not follow from one coarse statistic
Lower-scale experiential difference	C1-OI; lower-scale ontic nonisomorphism	requires independently identified lower-scale actual tokens
Restricted exclusive-selection obstruction	symmetry orbit; equivariant selector; overlap; nonempty exclusive-output requirement	pure selection theorem; independent of C1/U1
E1 No First Conscious Ancestor	U1; comparison-family members actually instantiated as valid tokens	no continuity premise is required for experience existence
E2 Fine-Grained Structural Chains	C1; common quotient/geometry; epsilon-fine chain of actual tokens	does not require U1; transfers structural fineness, not experience existence
E3 Connected Existence Constancy	connected parameter space; continuity of binary experience-existence predicate; endpoint with $E = 1$	conditional pure-mathematical lemma; independent of C1/U1/U2

Result / node	Required premises	Scope boundary
Strict evolutionary continuity application Evolutionary synthesis	U2; independently justified physical path continuity E1 plus either E2, or U2 + physical path continuity	application, not proof of C1 nonempty throughout + changing organization; not a scalar monotone ladder direct identity consequence, not an independent empirical law independent of C1/U1
Structural transformation	C1-OI; physical structural nonisomorphism	
Three-register/four-setting mathematics Toy whole-token experiential divergence	stipulated finite dynamics, boundaries, interventions exact whole-model divergence; actuality/completeness; C1-OI	conditional interpretation; no measured phenomenal difference
Toy sub-process experiential invariance	exact sub-process invariance; actuality/completeness; C1-W	does not imply whole-network invariance
OSC one-way package	C1; one-way bridge; measurement assumptions	operational implication comes from the bridge; success does not independently prove C1
OSC two-way package	C1; one-way bridge; reflection bridge; measurement assumptions	reflection must be separately justified
PTVSP	preregistration/freeze methodology	methodological safeguard; independent of C1 truth
Finite bridge failure	frozen package; mismatch; error model	challenges the operational package first; does not automatically isolate C1
UCT-II formal reconstruction	frozen physical view; BAC/TFR; theory-specific conventions	independent of C1/U1 as formal reconstruction
UCT-II no-gate relocation	U1; scoped actual nonuniversality witness	distinct from reconstruction
UCT-II experiential interpretation	C1; reconstructed/identified physical organization	distinct from reconstruction and empirical validation

C.1 Formal-DAG audit

The v1.2 dependency graph contains **78 nodes** and **105 directed edges**. It is a directed acyclic graph with:

$$\text{cycles} = 0.$$

Its transitive reduction contains **97 edges**.

The following checks pass:

1. no path from finite views, estimates, subjecthood, phenomenal targets, bridge packages, PTVSP, toy-model mathematics, evolutionary-path premises, E3, or UCT-II reconstruction back into C1/U1/U2;
2. U1 is not upstream of C1;
3. U2 is not upstream of C1 or U1;
4. E1 depends on U1 but not U2;
5. E2 depends on C1 and structural geometry but not U1;

6. E3 is independent of C1/U1/U2;
7. the symmetry-orbit selection obstruction is independent of C1/U1;
8. toy-model mathematics is independent of C1/U1;
9. UCT-II formal reconstruction is independent of C1/U1.

Graph correctness verifies dependency direction only. It does not prove C1, ontology completeness, bridge validity, metric choice, or empirical truth.

C.2 Conceptual firewall map

actual token $\neq \mathbf{D}^{ontic} \neq D_v \neq \widehat{D}_v \neq \text{subject.}$

experience existence $\neq$ experience organization $\neq$ report/access/self-model.

toy-model mathematics $\neq$ phenomenal validation.

formal reconstruction $\neq$ no-gate relocation $\neq$ experiential interpretation.

evolutionary continuity $\neq$ proof of C1.

A machine-readable graph and edge ledger are supplied separately in the v1.2 prepublication audit package.

Appendix D. Optional checks on a finite coarse description

These checks concern a proposed scientific model, not the existence of experience. Actual processes need not admit autonomous, closed coarse descriptions. The finite process criterion in Section 2.5 does not depend on the checks below.

A useful model dossier identifies the actual referent and its physical state carriers or distributed organization; declares the time horizon, boundary, and omitted feedback; and distinguishes implemented relations from a calculated summary. Predictive success alone does not create a new token. A literal register is not required for a distributed realization.

For a deterministic lower-level update $x^+ = T_e(x)$ and a surjective coarse map $\pi : X \rightarrow M$, a closed deterministic update $\bar{T}_e$ exists exactly when

$$\pi(x) = \pi(x') \implies \pi(T_e x) = \pi(T_e x').$$

Proof. If $\pi T_e = \bar{T}_e \pi$, equal coarse states have equal coarse successors. Conversely, define $\bar{T}_e(m) = \pi(T_e x)$ using any representative of m . The displayed condition makes the definition representative-independent, and surjectivity supplies a representative. $\square$

For a declared state-reset intervention I , an additional intervention correspondence ω can be checked by

$$\pi(T_e(Ix)) = \bar{T}_e(\omega(I)\pi(x)).$$

The reset is applied before the transition. Mechanism-changing interventions require separately indexed update maps; the reset equation does not automatically cover them. Approximate, probabilistic, or history-dependent models need their own error and sufficiency

conditions. Failure of a closed deterministic description is a failure of that model, not absence of experience in the actual process.

This mathematics belongs to established lumpability and causal-abstraction methodology (Buchholz, 1994; Rubenstein et al., 2017; Beckers & Halpern, 2019). UCT claims no mathematical priority. Appendix B supplies both an implemented closed process and examples illustrating why a closed calculated feature is not automatically another token, and why an arbitrary feature can fail closure. These distinctions keep optional modeling requirements separate from the common explanatory principle for elementary, cellular, and human experience.